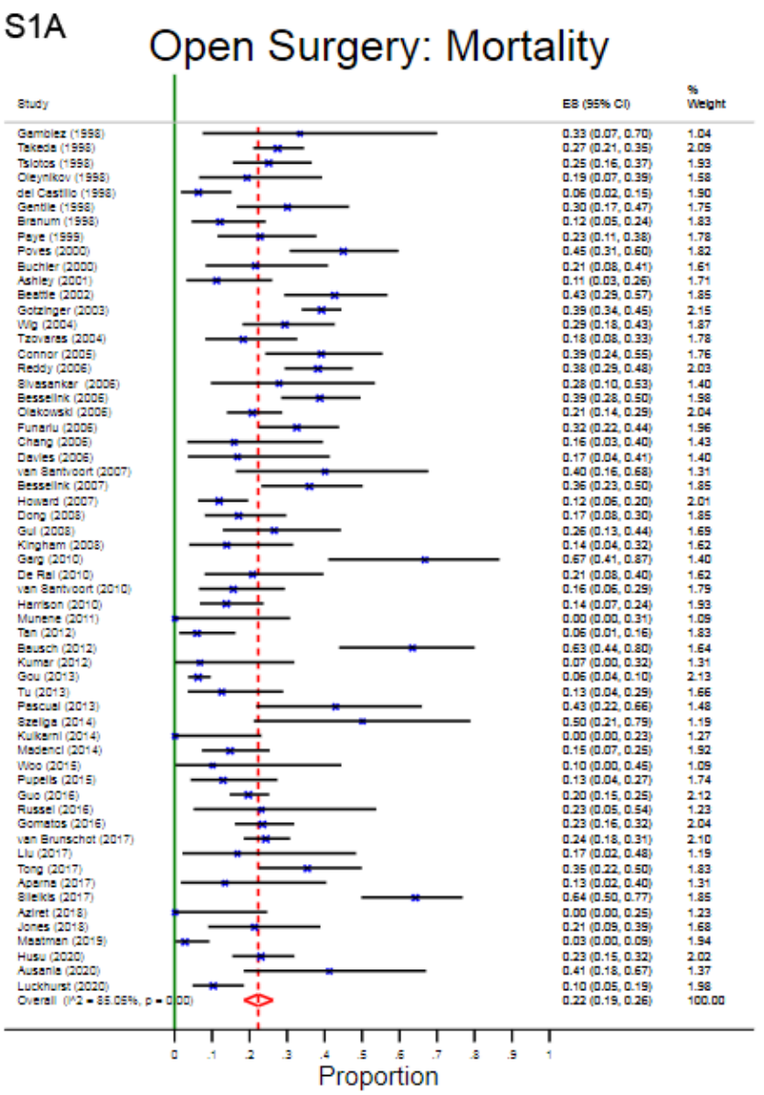
Supplementary Material

Figure S1. Forest plot of the pooled prevalence of mortality rate after drainage procedures in necrotizing pancreatitis. A. Open surgery; B. Minimally invasive surgery; C. Step up approach; D. Endoscopic drainage

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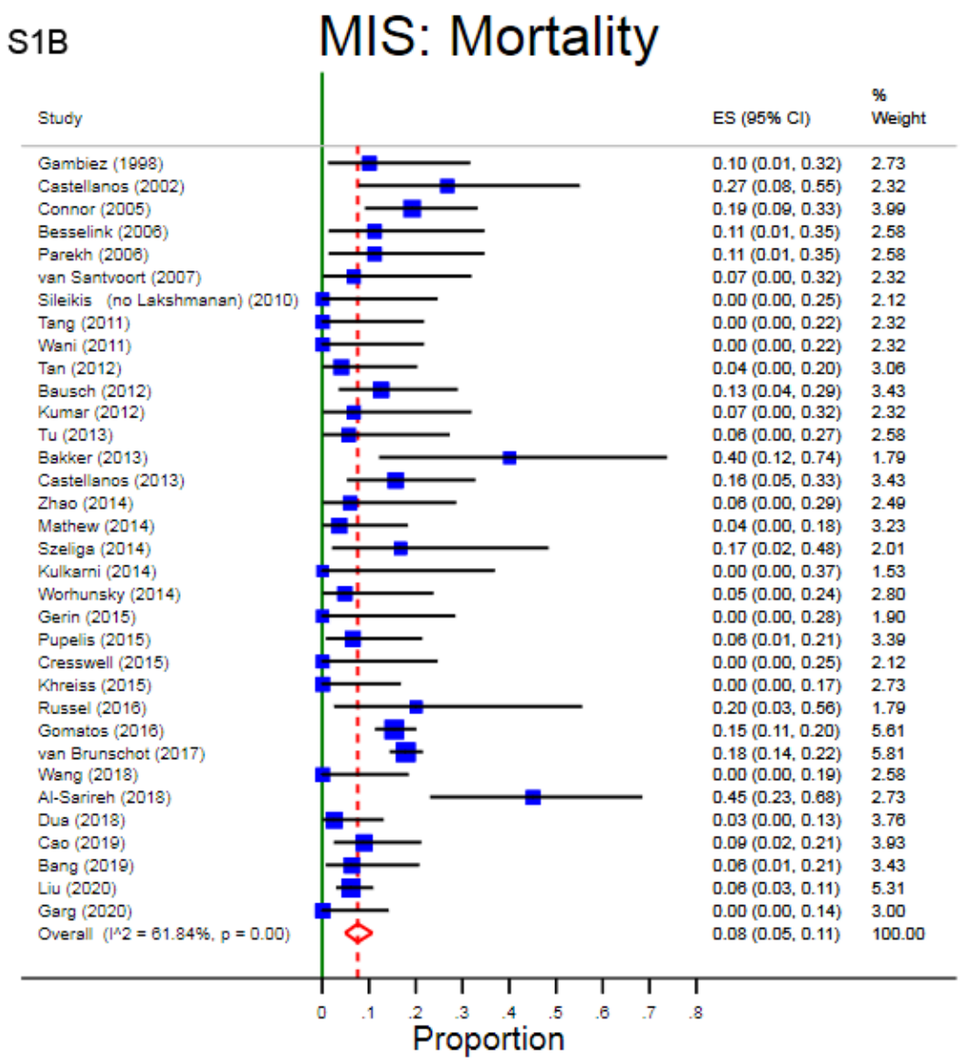
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S1A



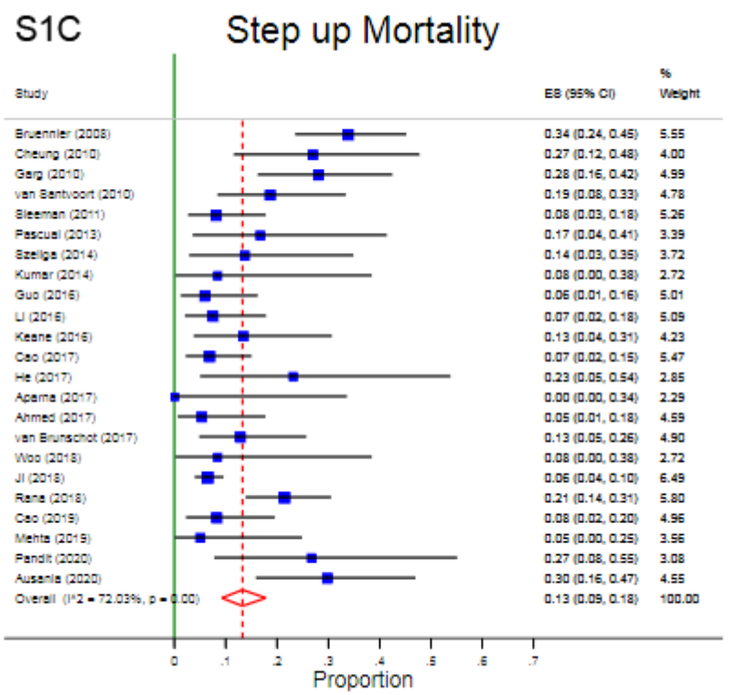
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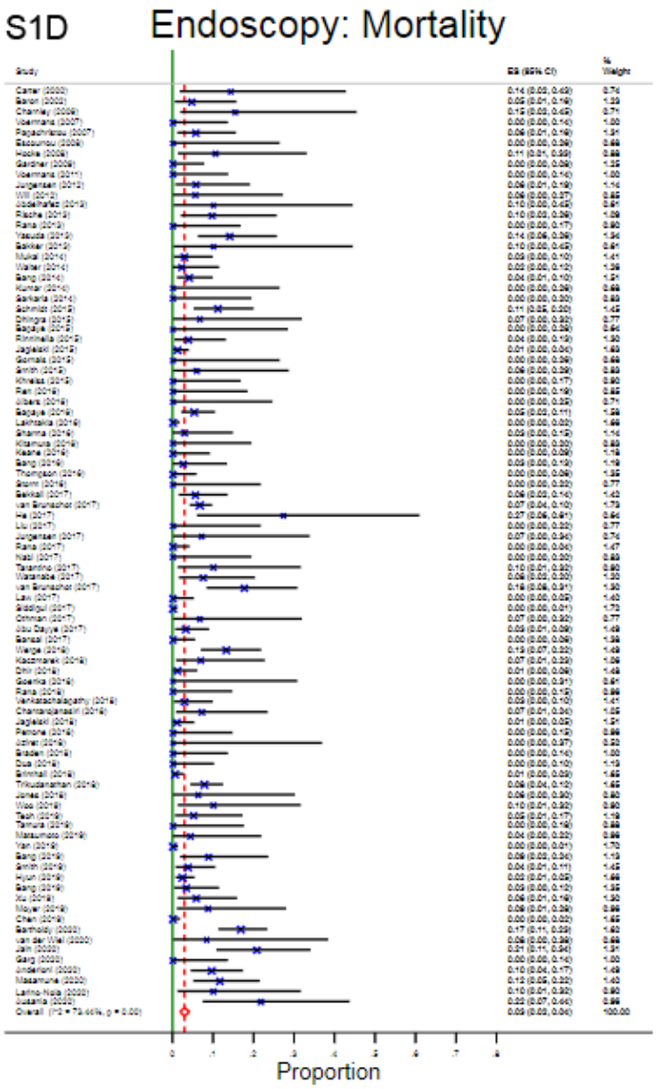


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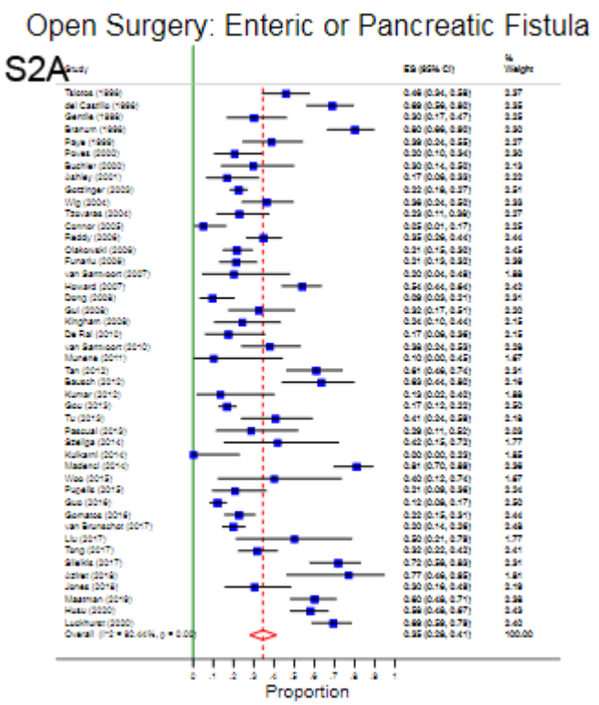


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Supplementary material

Figure S2. Forest plot of the pooled prevalence of pancreatic or enteric fistula after drainage procedures in necrotizing pancreatitis.

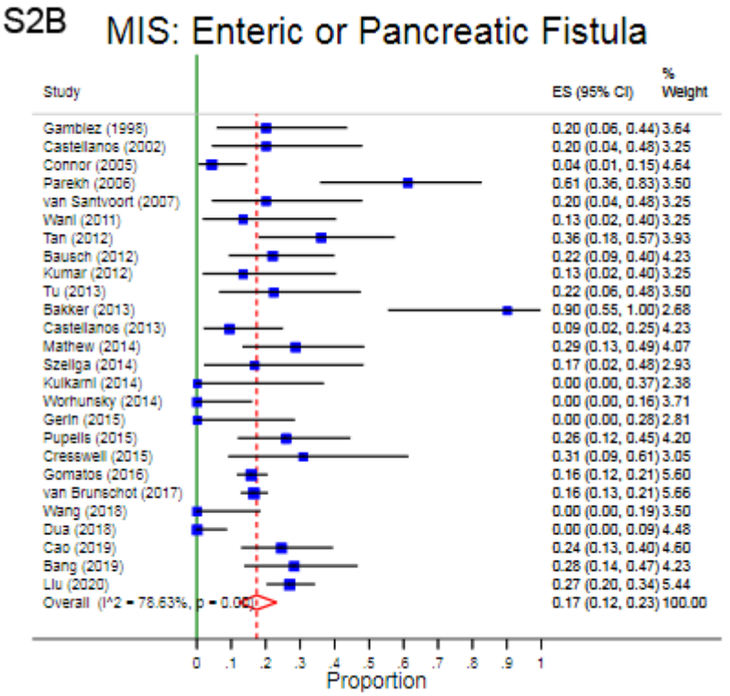
A. Open surgery; B. Minimally invasive surgery; C. Step up approach; D. Endoscopic drainage



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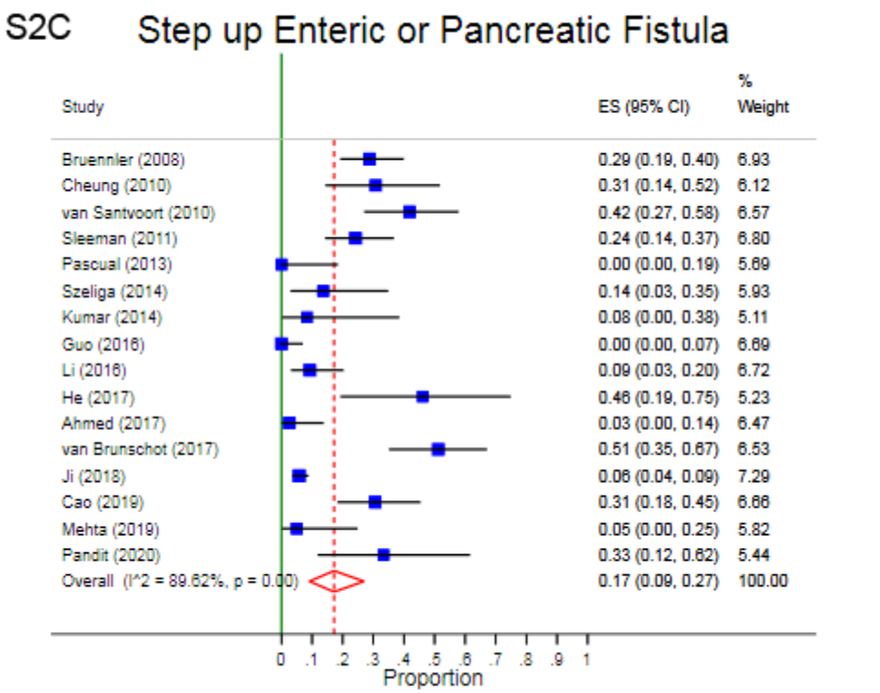
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S2B



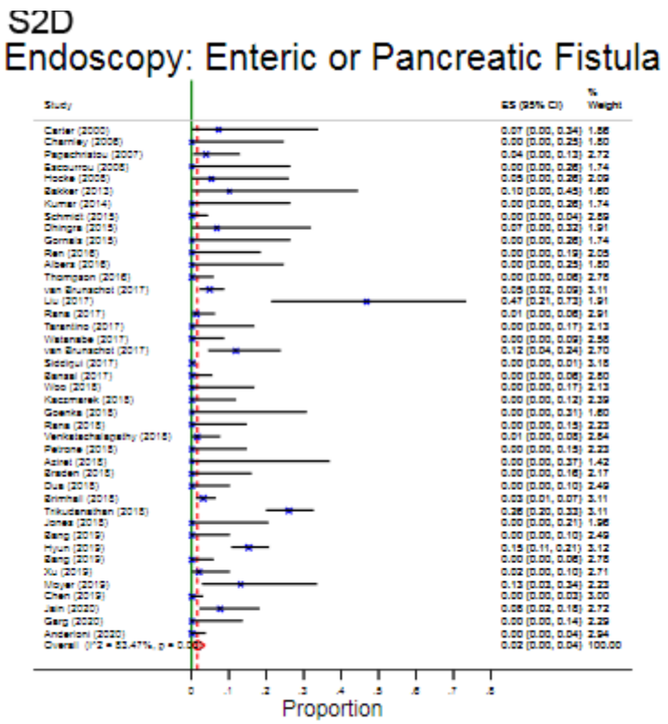
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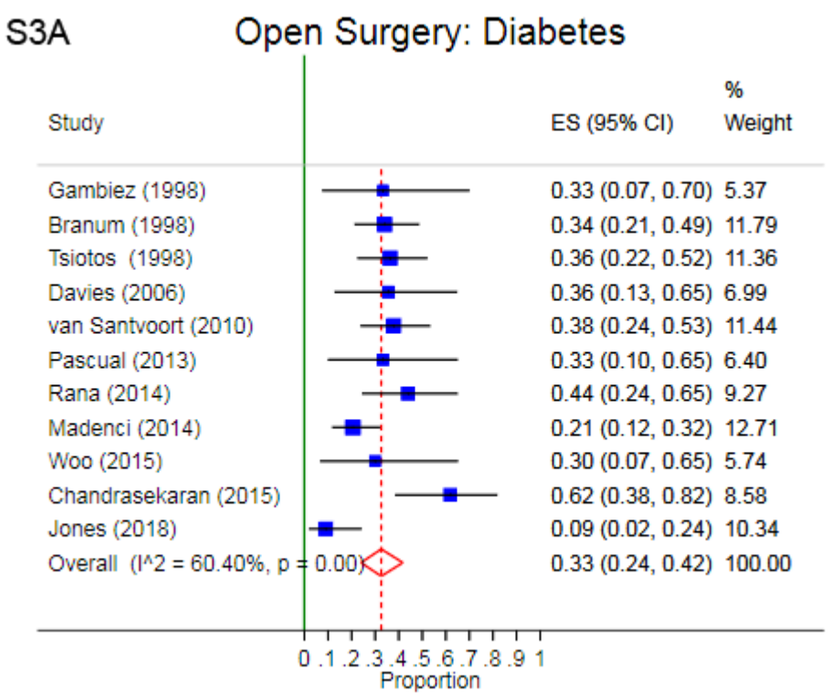
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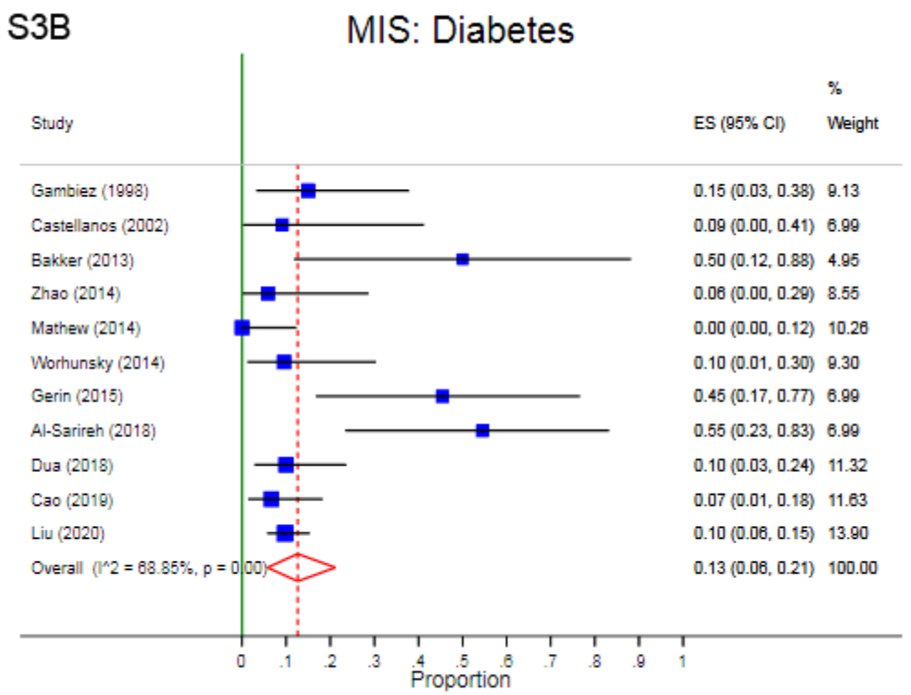
Supplementary material

Figure S3. Forest plot of the pooled prevalence of new onset diabetes mellitus after drainage procedures in necrotizing pancreatitis. A. Open surgery; B. Minimally invasive surgery; C. Step up approach; D. Endoscopic drainage



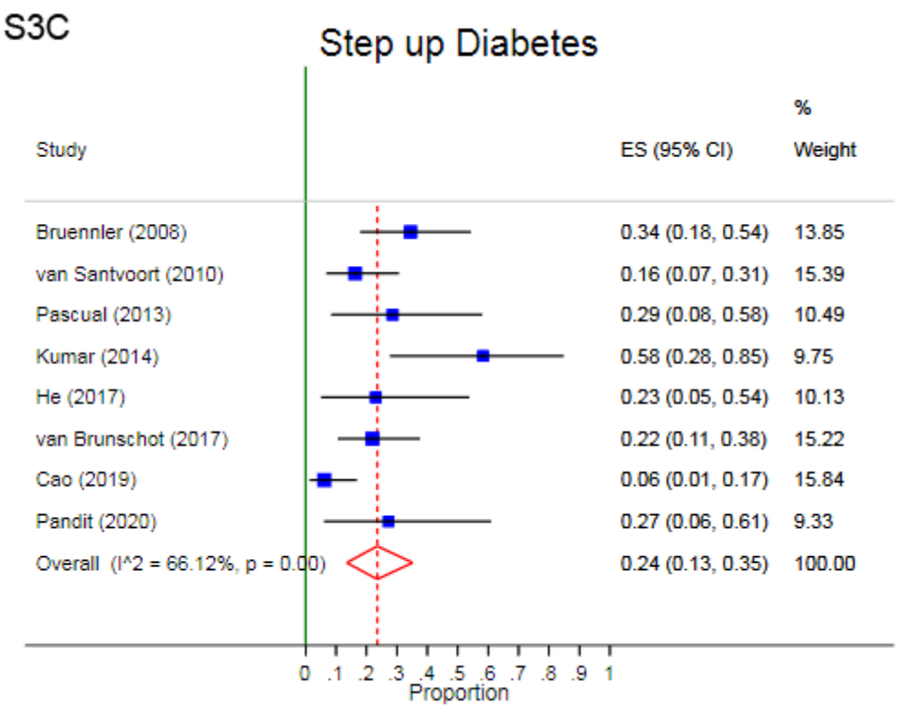
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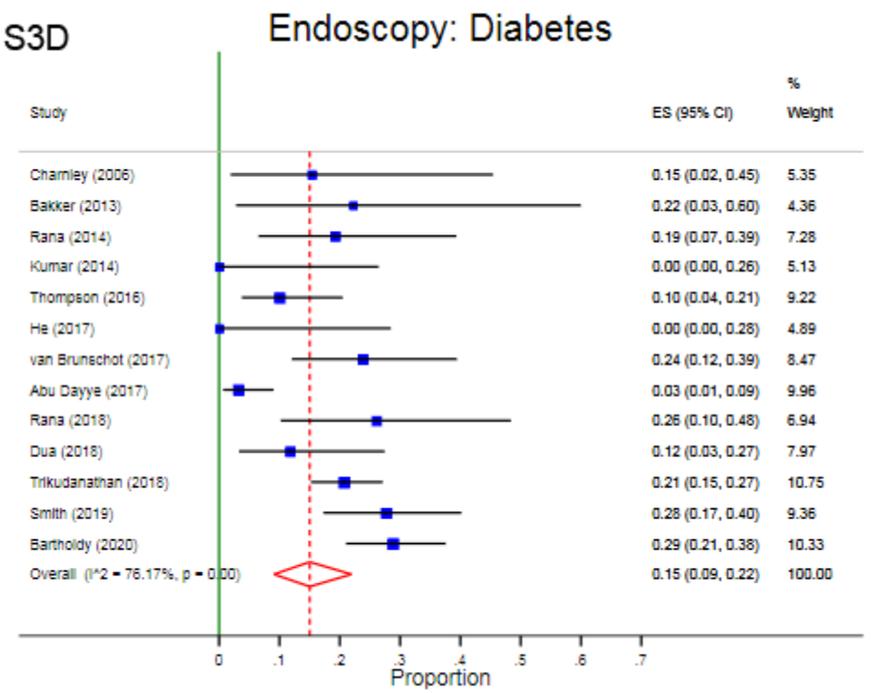
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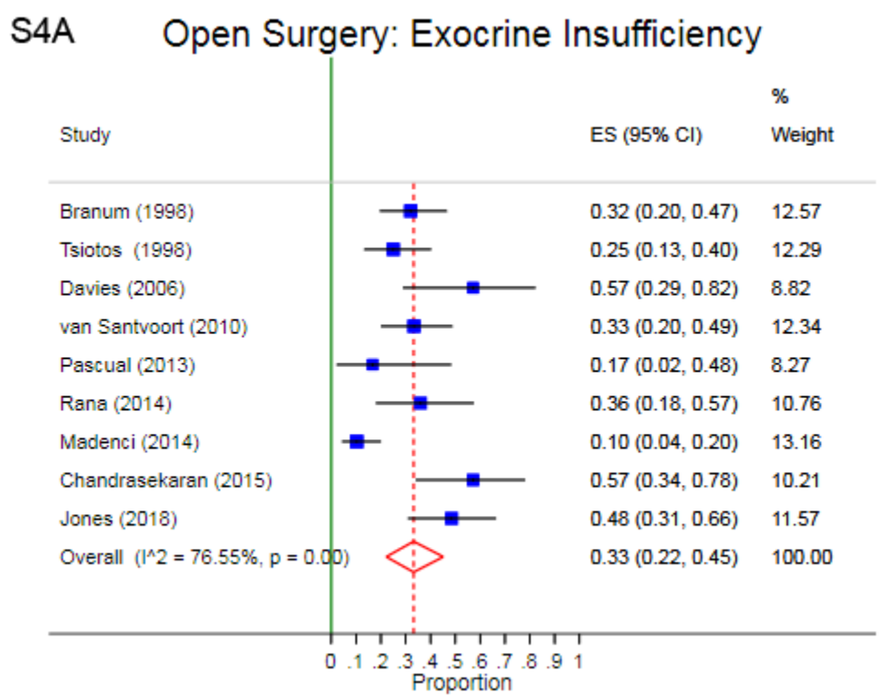
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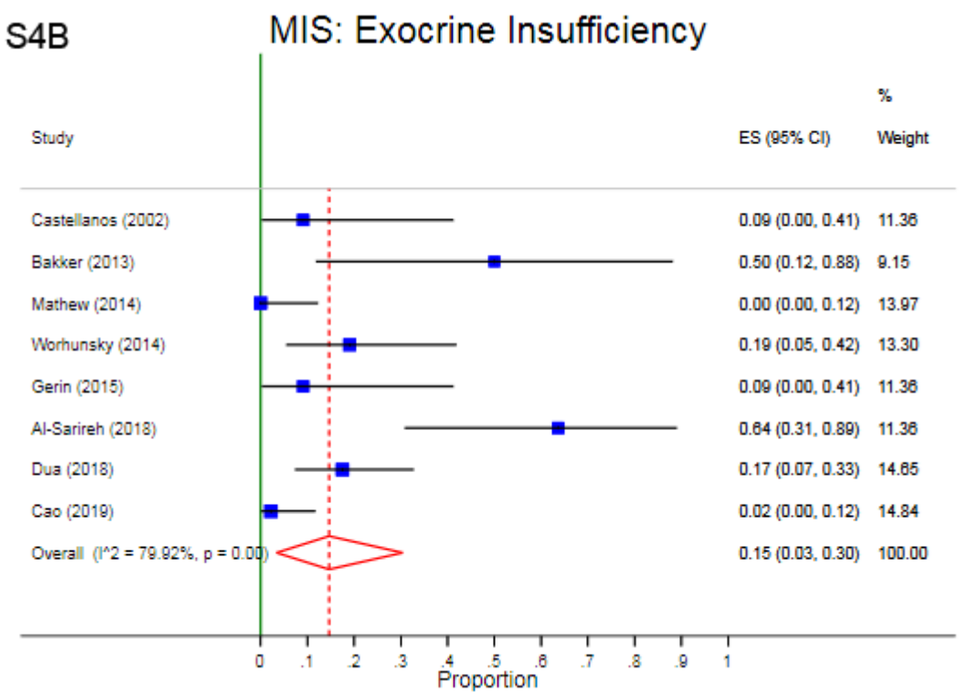
Supplementary material

Figure S4. Forest plot of the pooled prevalence of pancreatic insufficiency after drainage procedures in necrotizing pancreatitis. A. Open surgery; B. Minimally invasive surgery; C. Step up approach; D. Endoscopic drainage



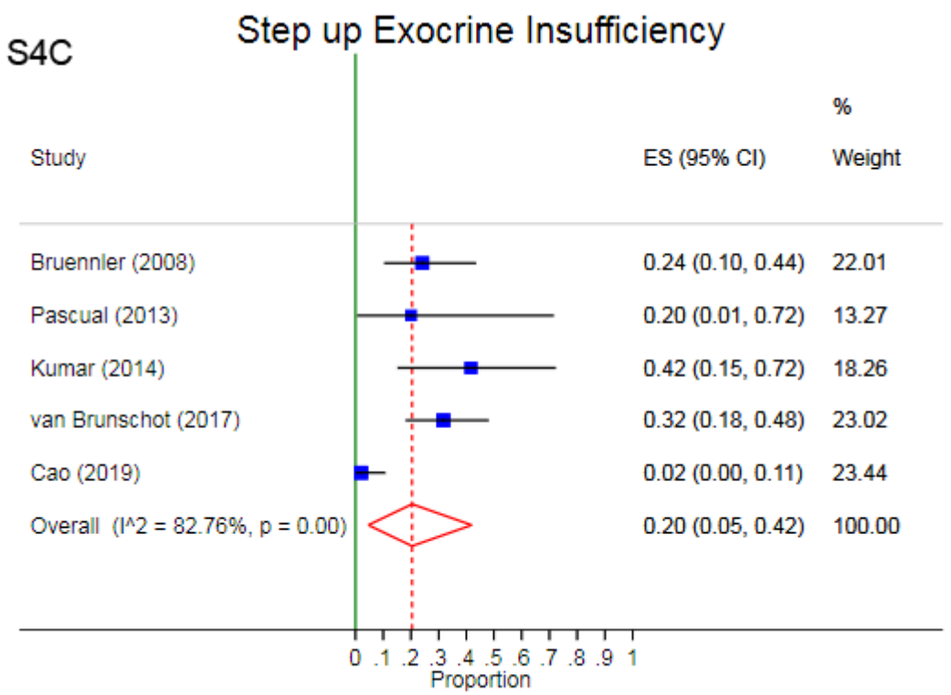
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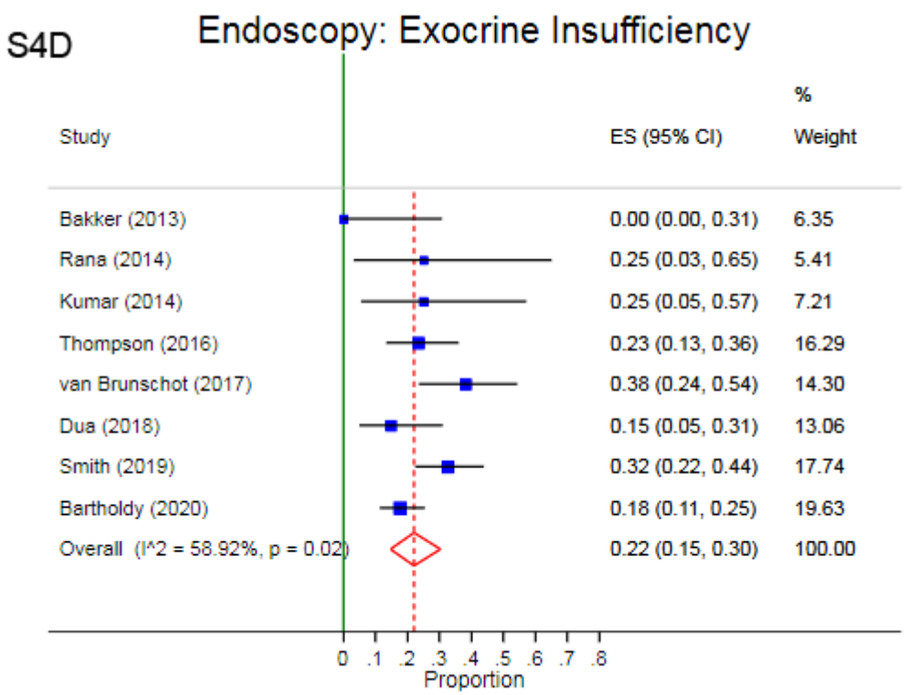
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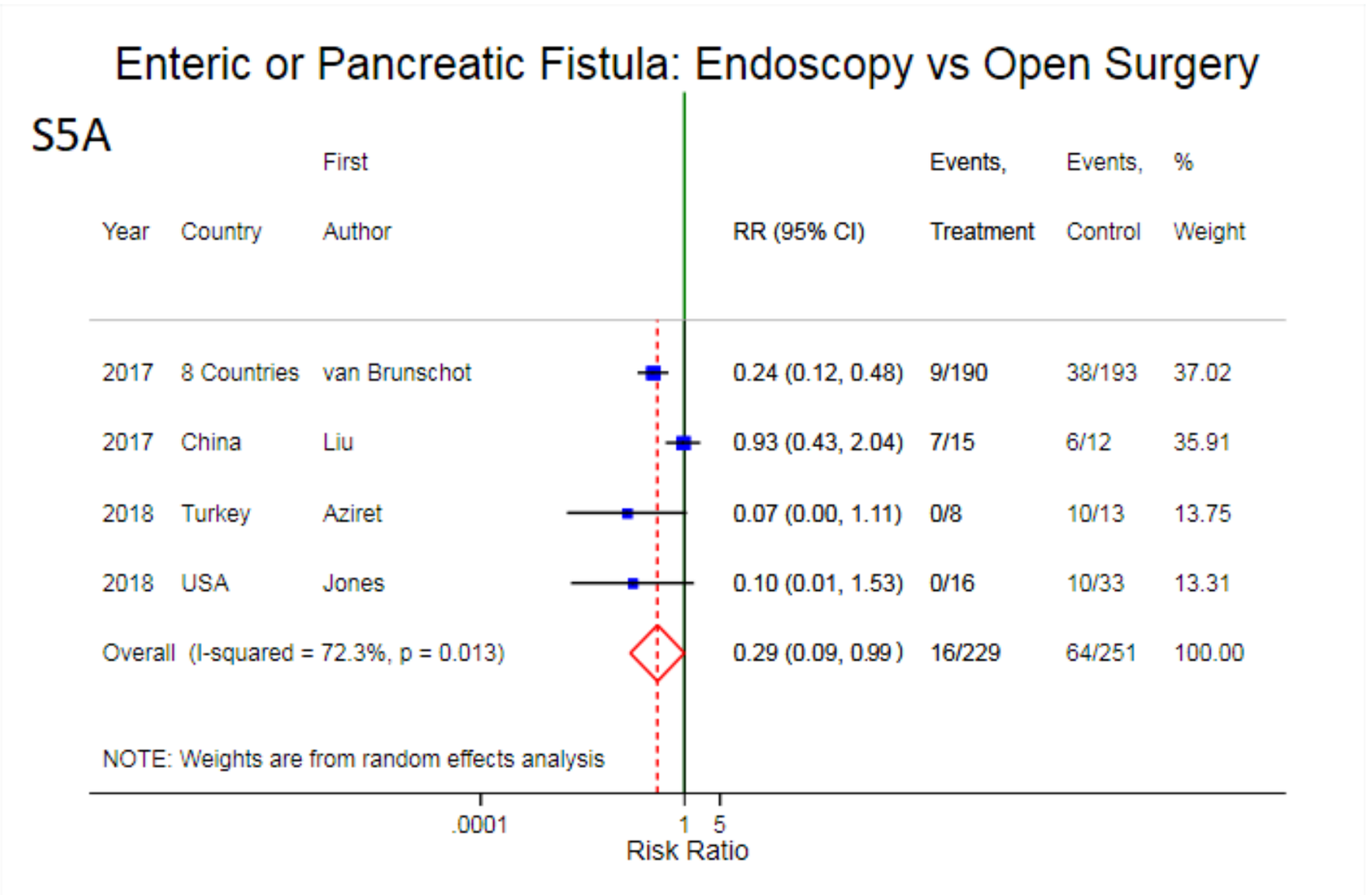
Supplementary material

Figure S5. Forest plot comparing fistula formation between endoscopic drainage and other types of drainage procedures in comparative studies.

A. Endoscopic drainage vs. open surgery; B. Endoscopic drainage vs. minimally invasive surgery; C. Endoscopic drainage vs. step up approach

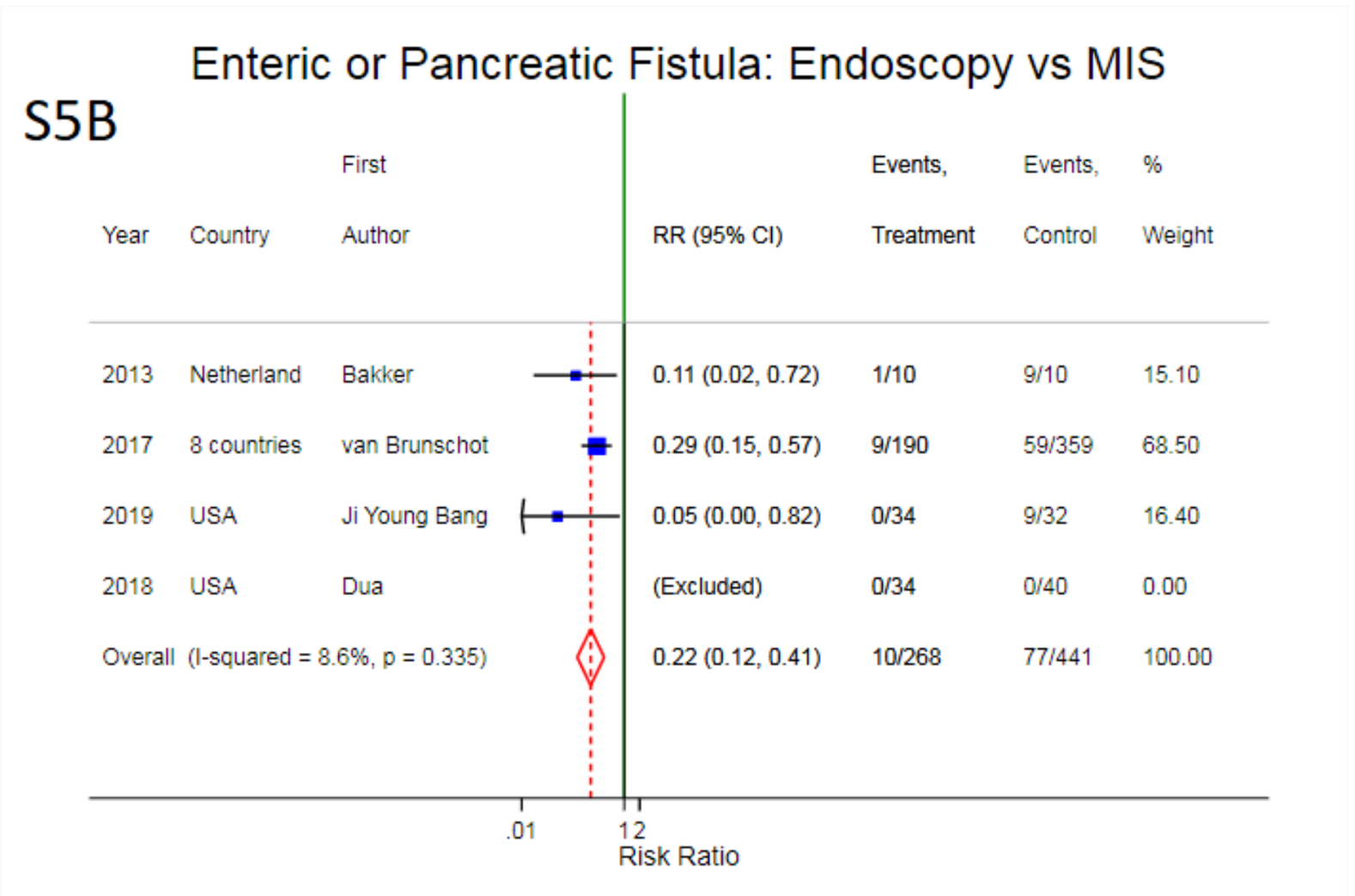
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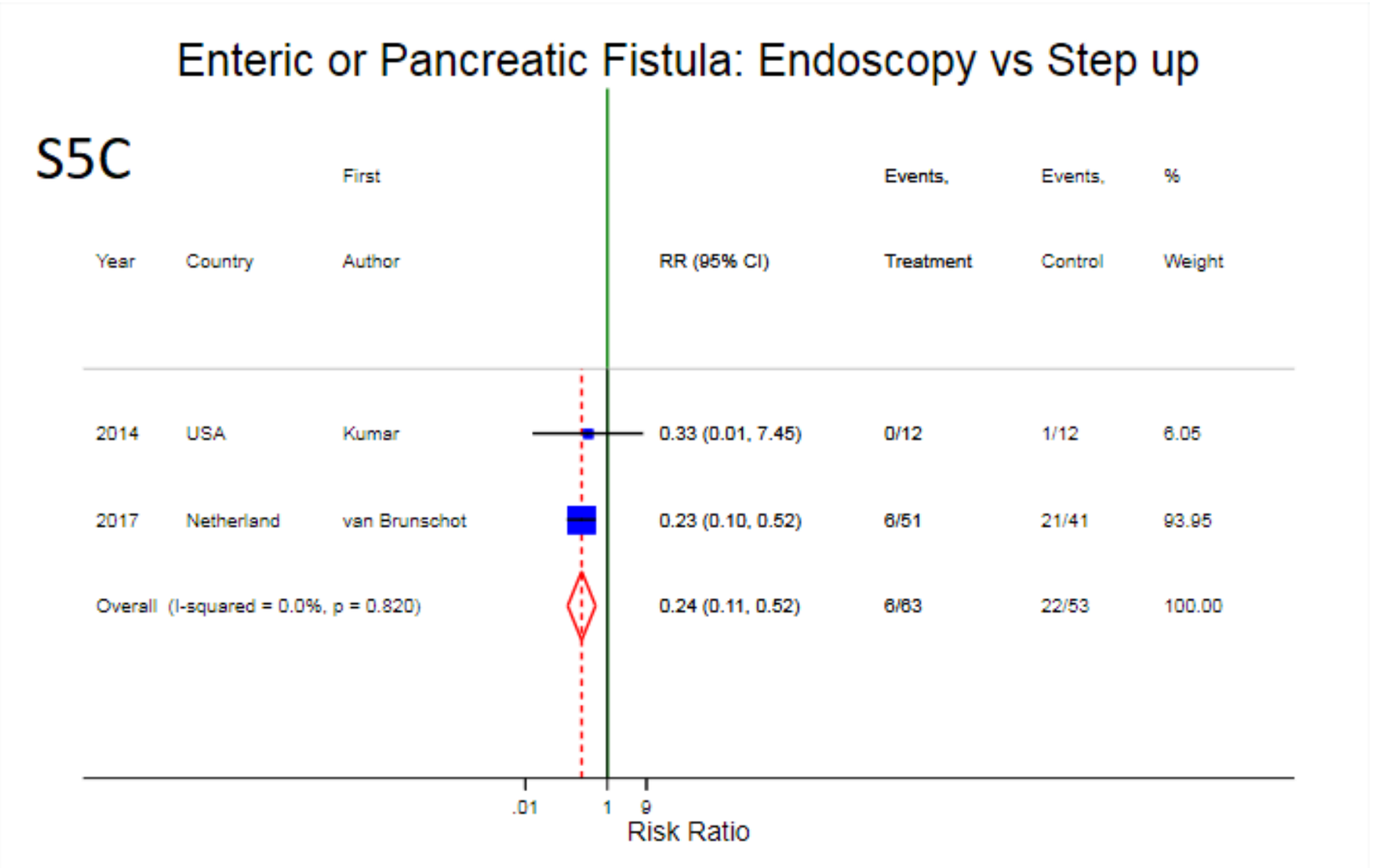


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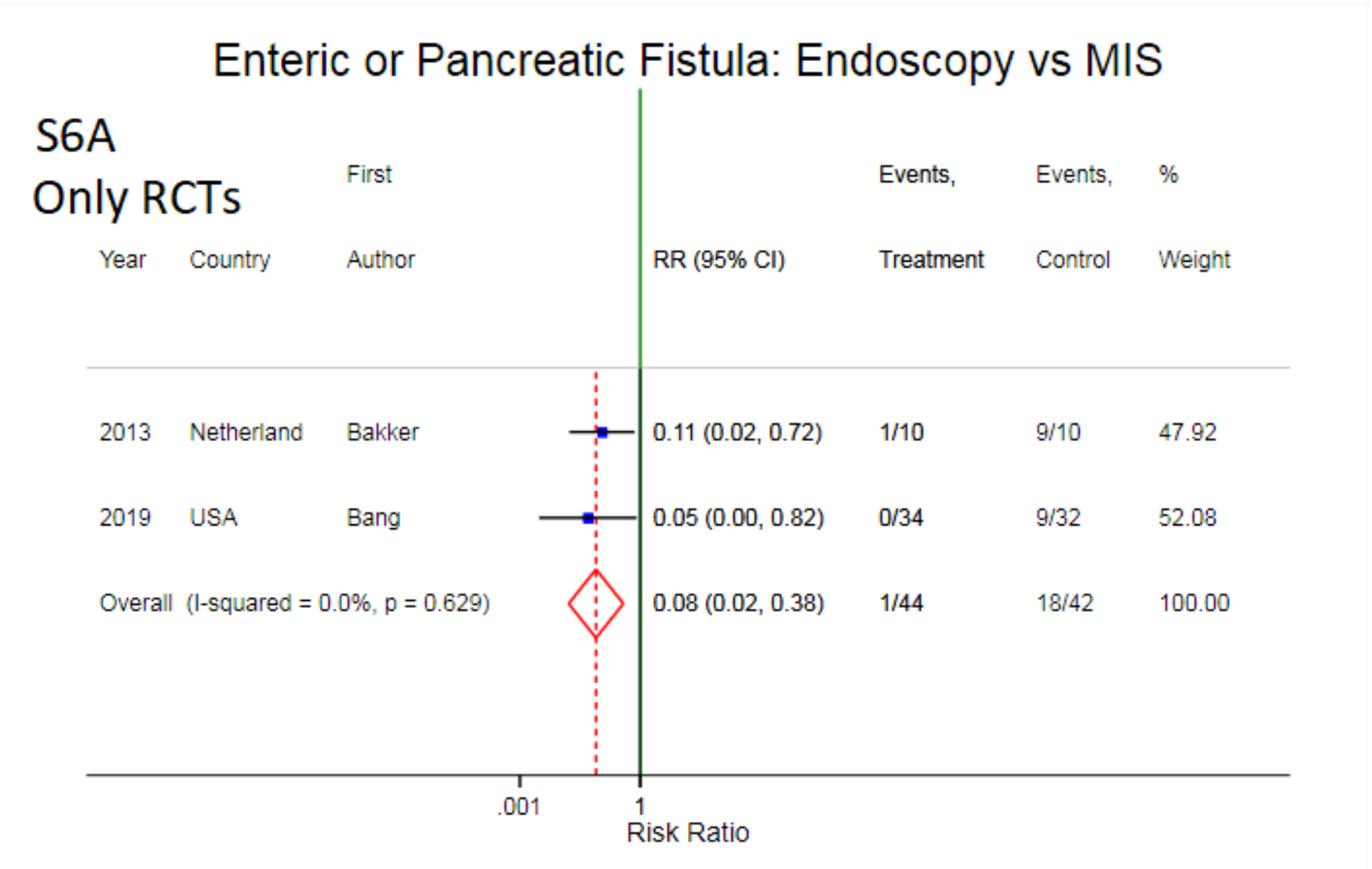
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Figure S6. Forest plot comparing of the outcome differences between endoscopic drainage and minimally invasive surgery in randomized controlled trials. A. Fistula formation; B. Mortality rate

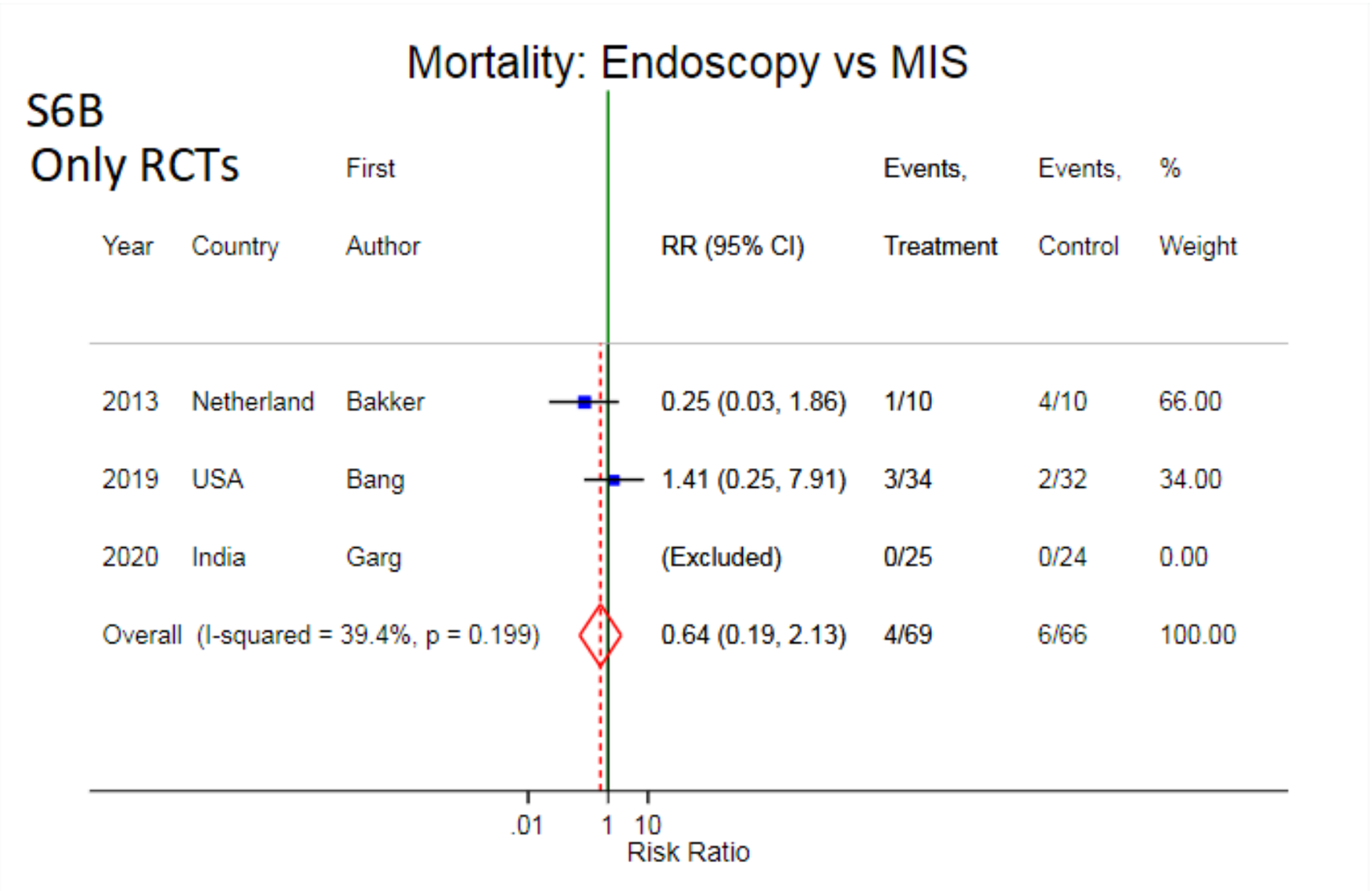
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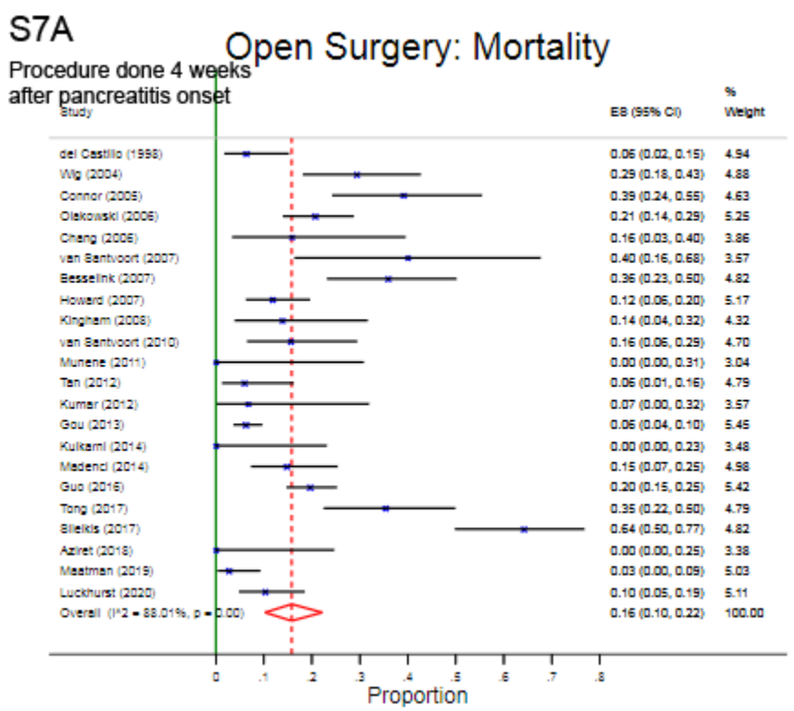
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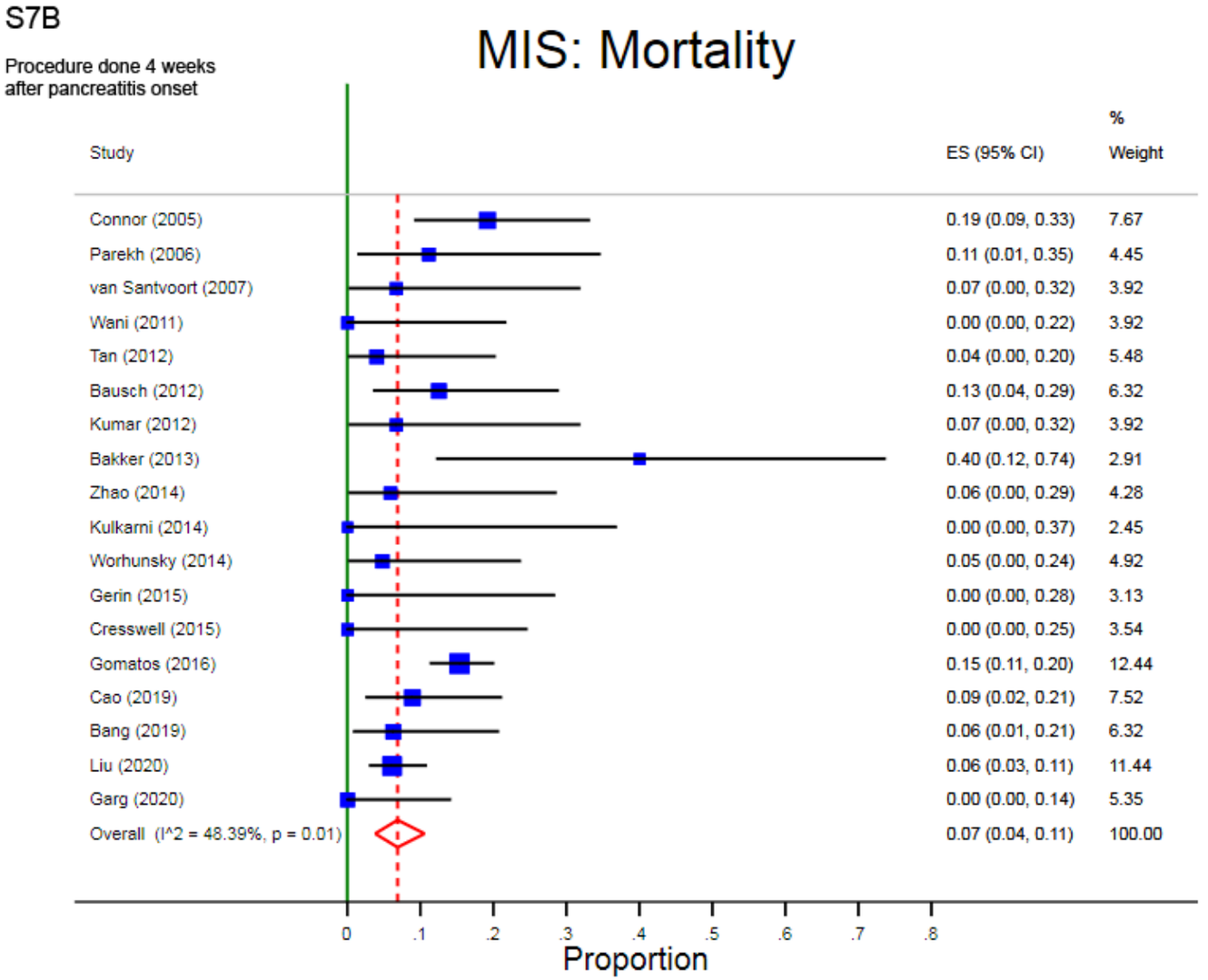
Figure S7. Forest plot of the pooled prevalence of mortality rate after drainage procedures in studies with the median time from pancreatitis onset to surgery of 4 weeks or more. A. Open surgery; B. Minimally invasive surgery; C. Step up approach; D. Endoscopic drainage



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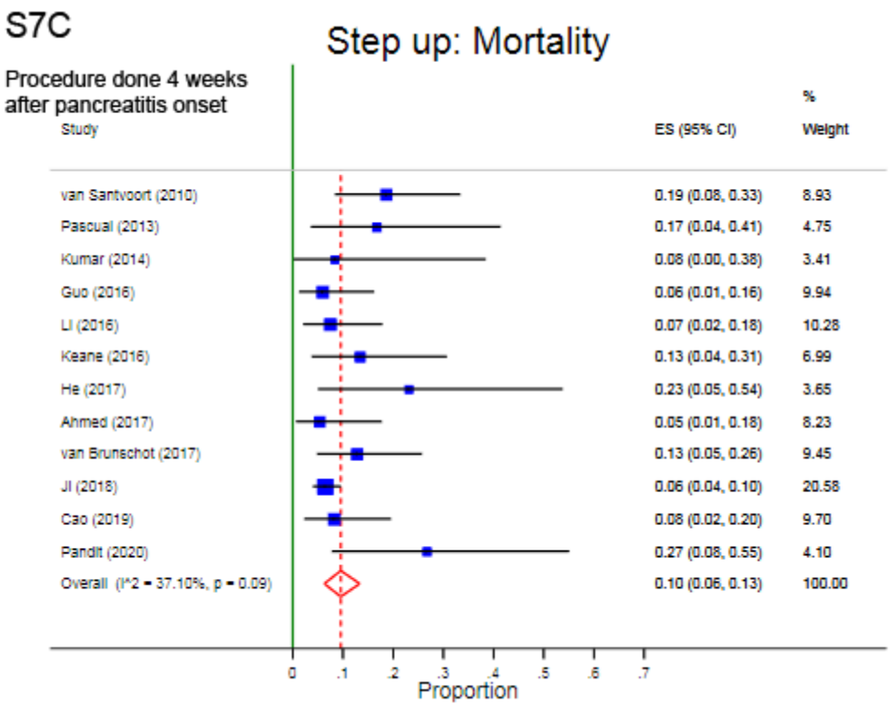
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S7B



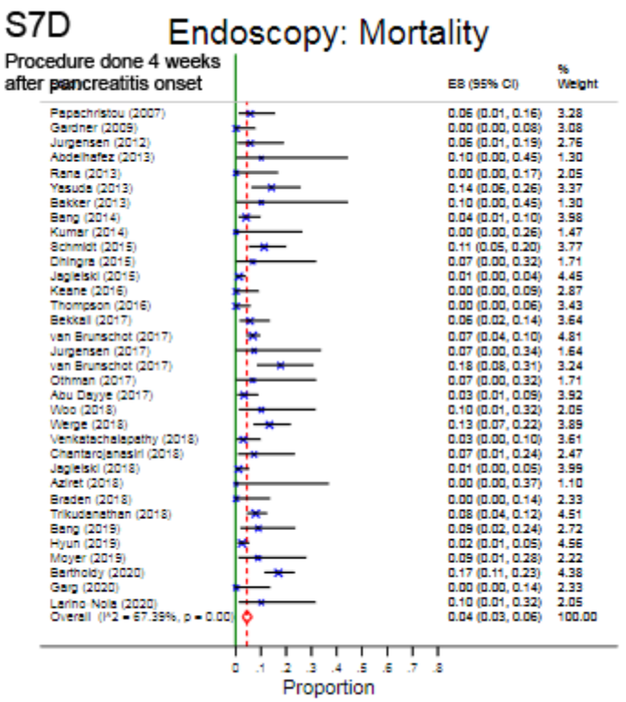
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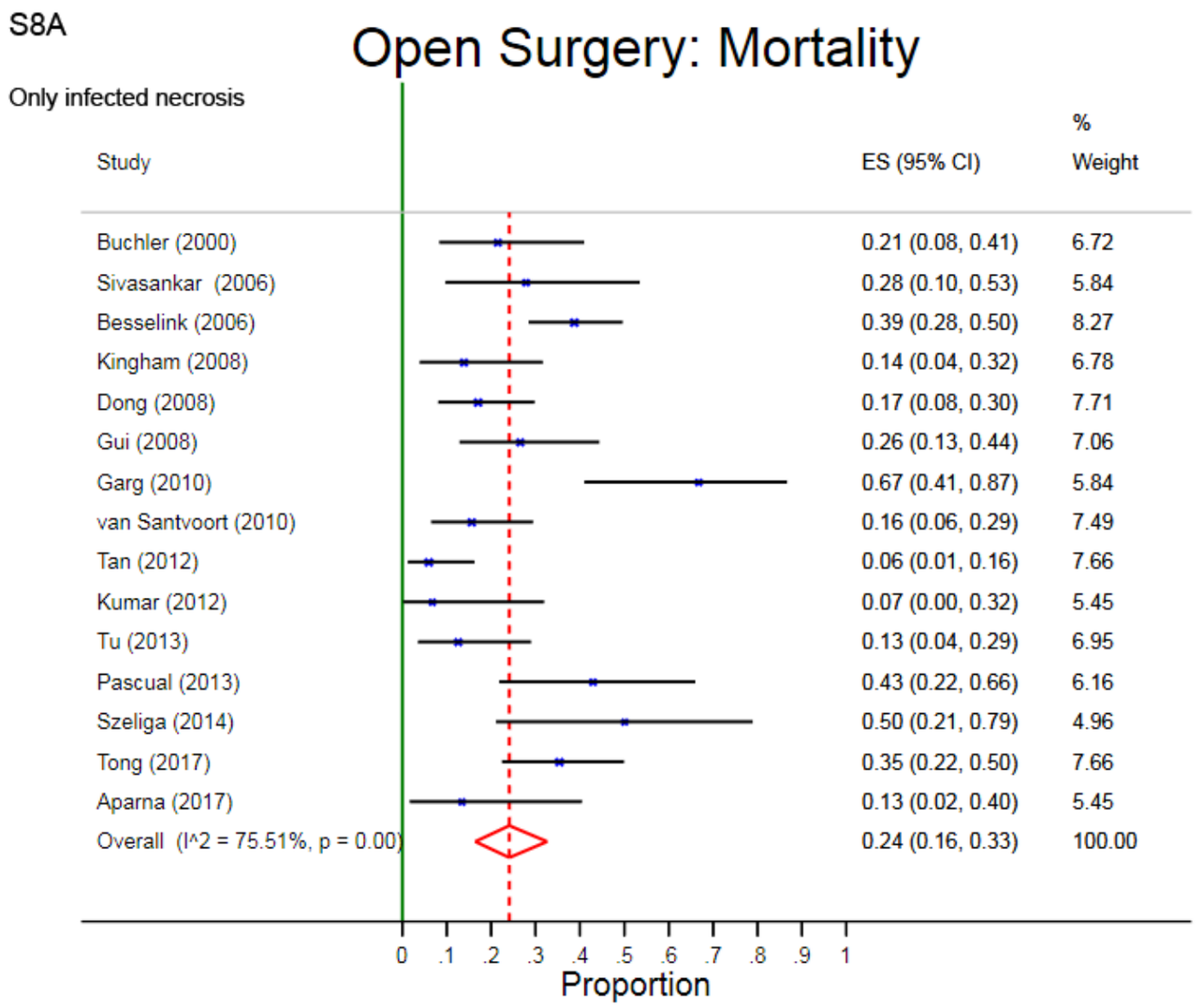
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Figure S8. Forest plot of the pooled prevalence of mortality rate after drainage procedures in studies only including patients with confirmed or suspected infected pancreatic necrosis. A. Open surgery; B. Minimally invasive surgery; C. Step up approach; D. Endoscopic drainage

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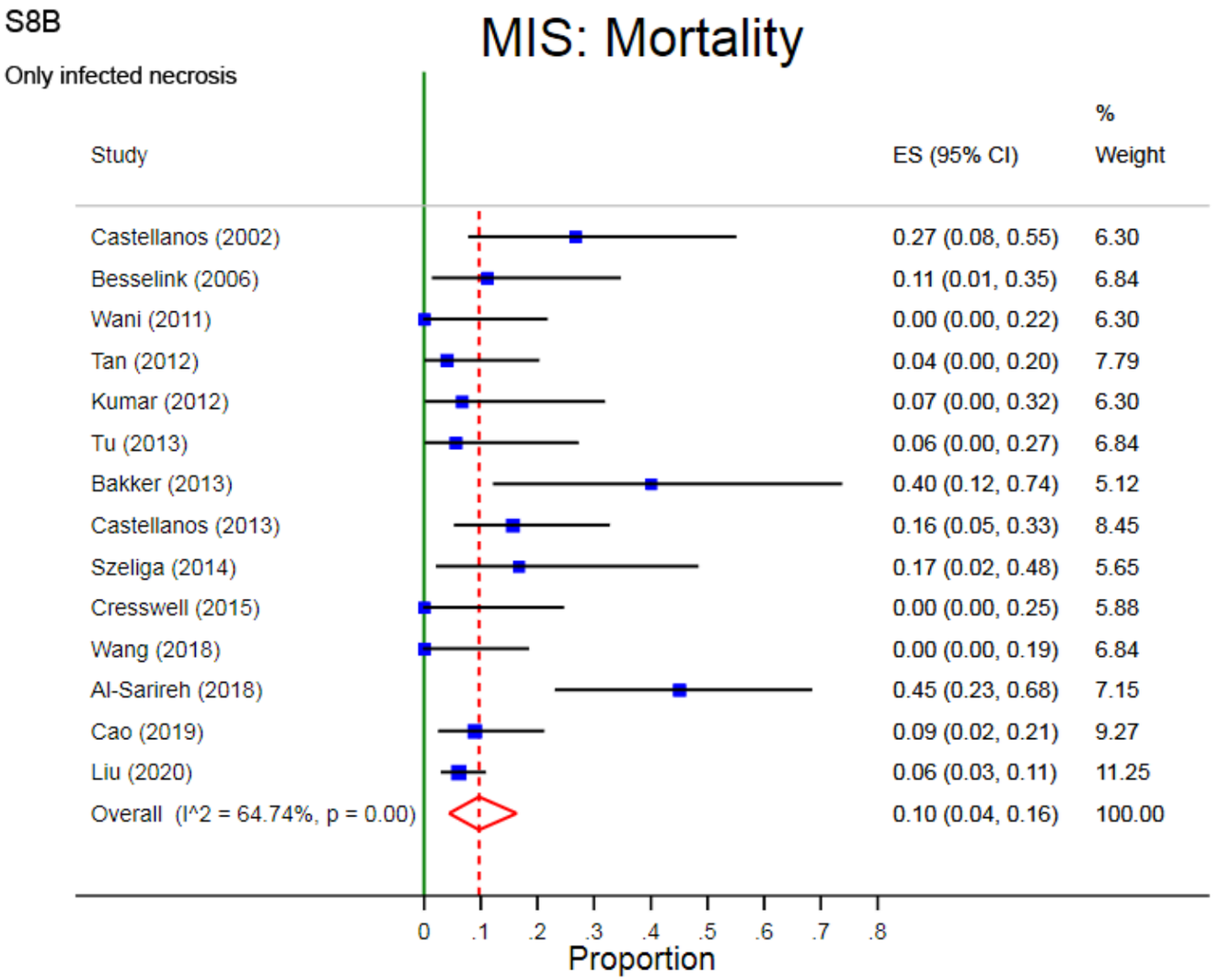
S8A



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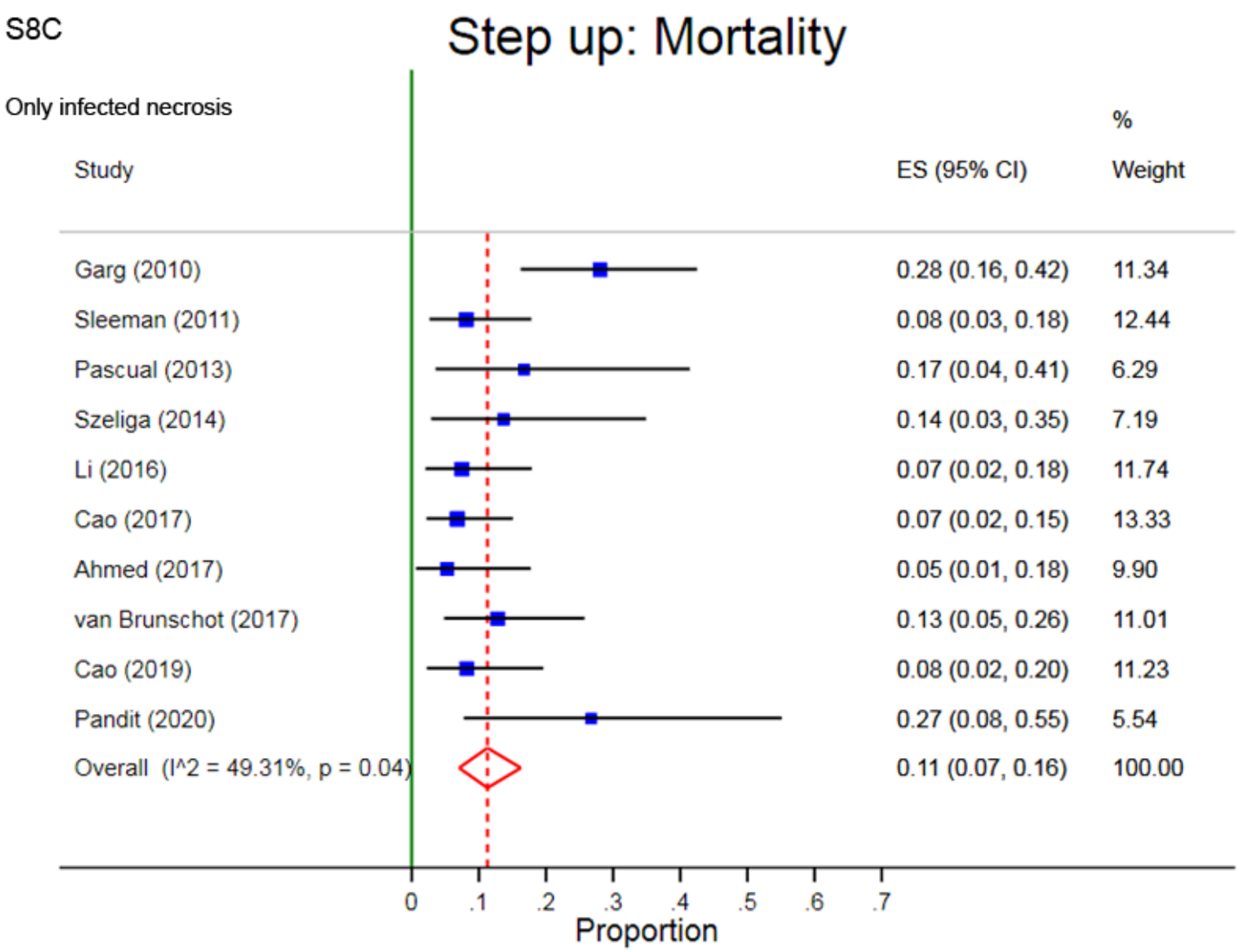
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S8B



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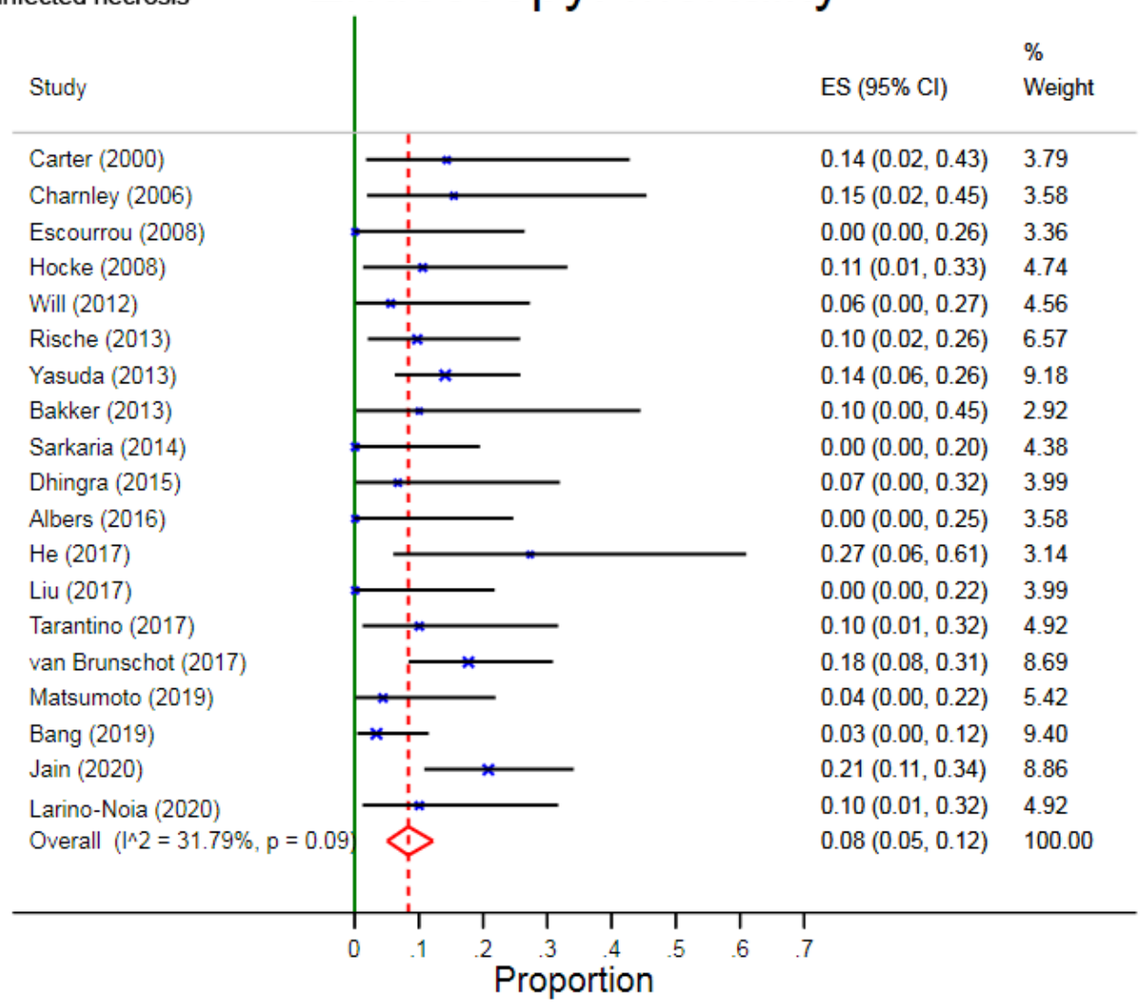


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S8D

Only infected necrosis



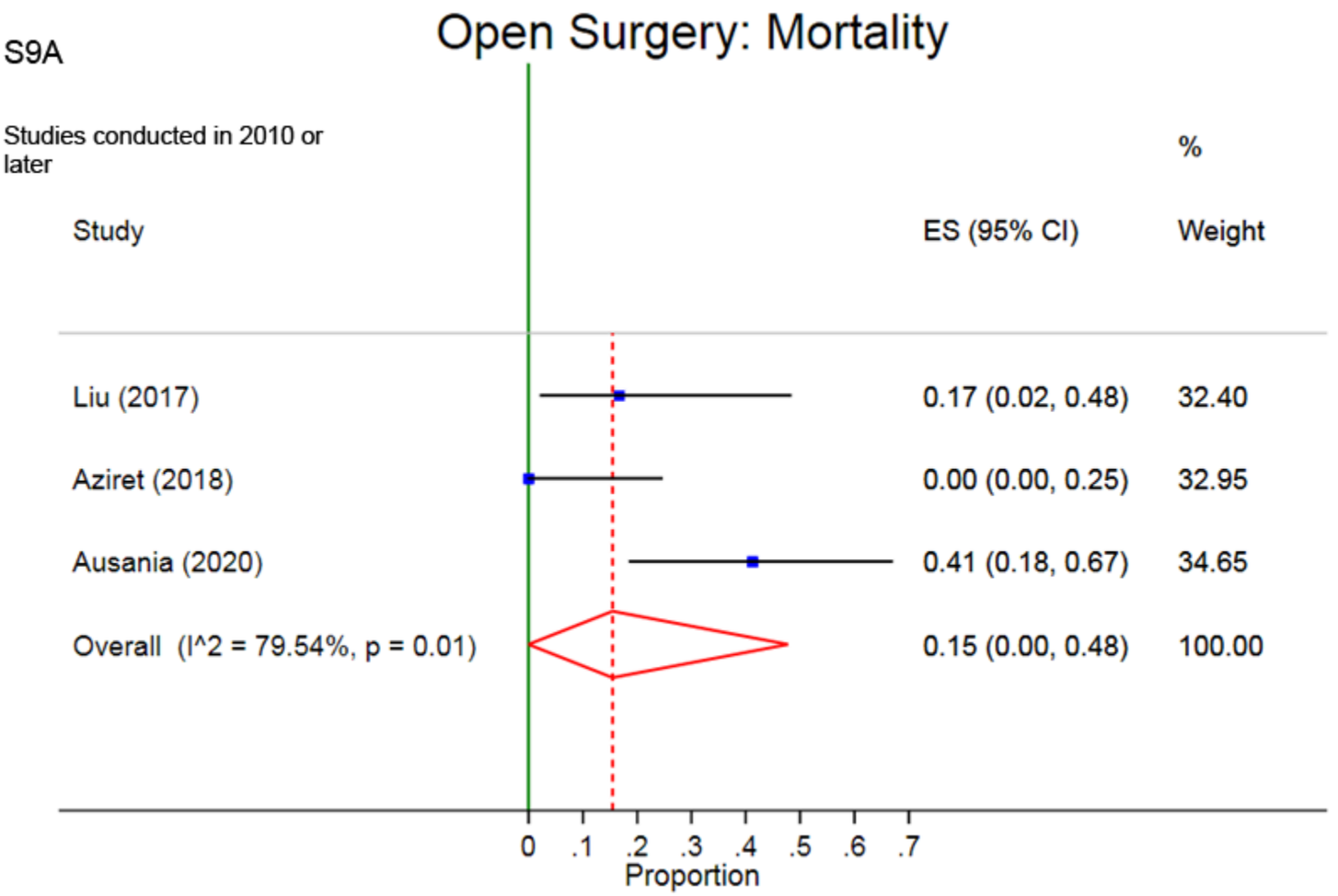
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Figure S9. Forest plot of the pooled prevalence of mortality rate after drainage procedures in articles with the study period limited to 2010 or later. A. Open surgery; B. Minimally invasive surgery; C. Step up approach; D. Endoscopic drainage

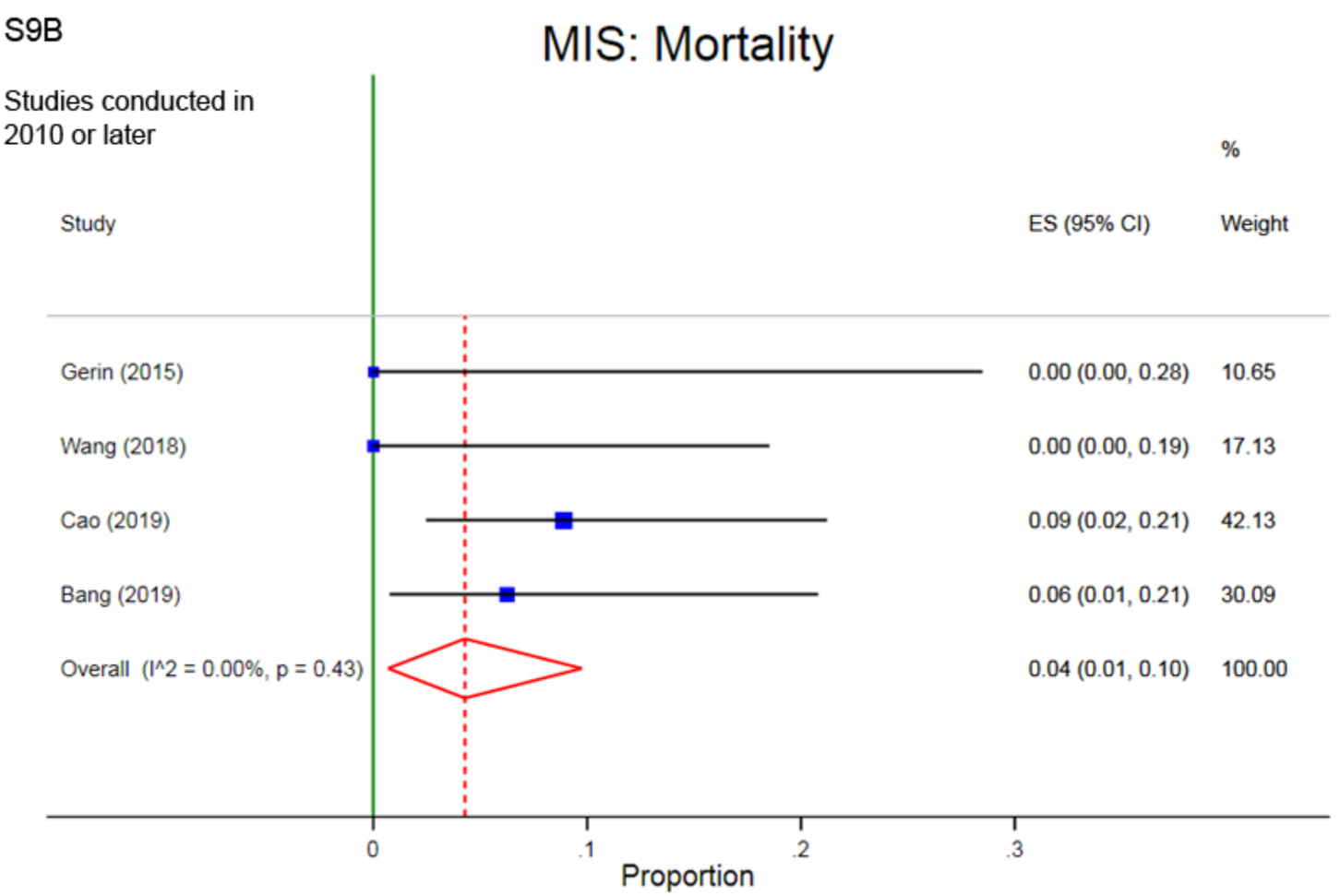
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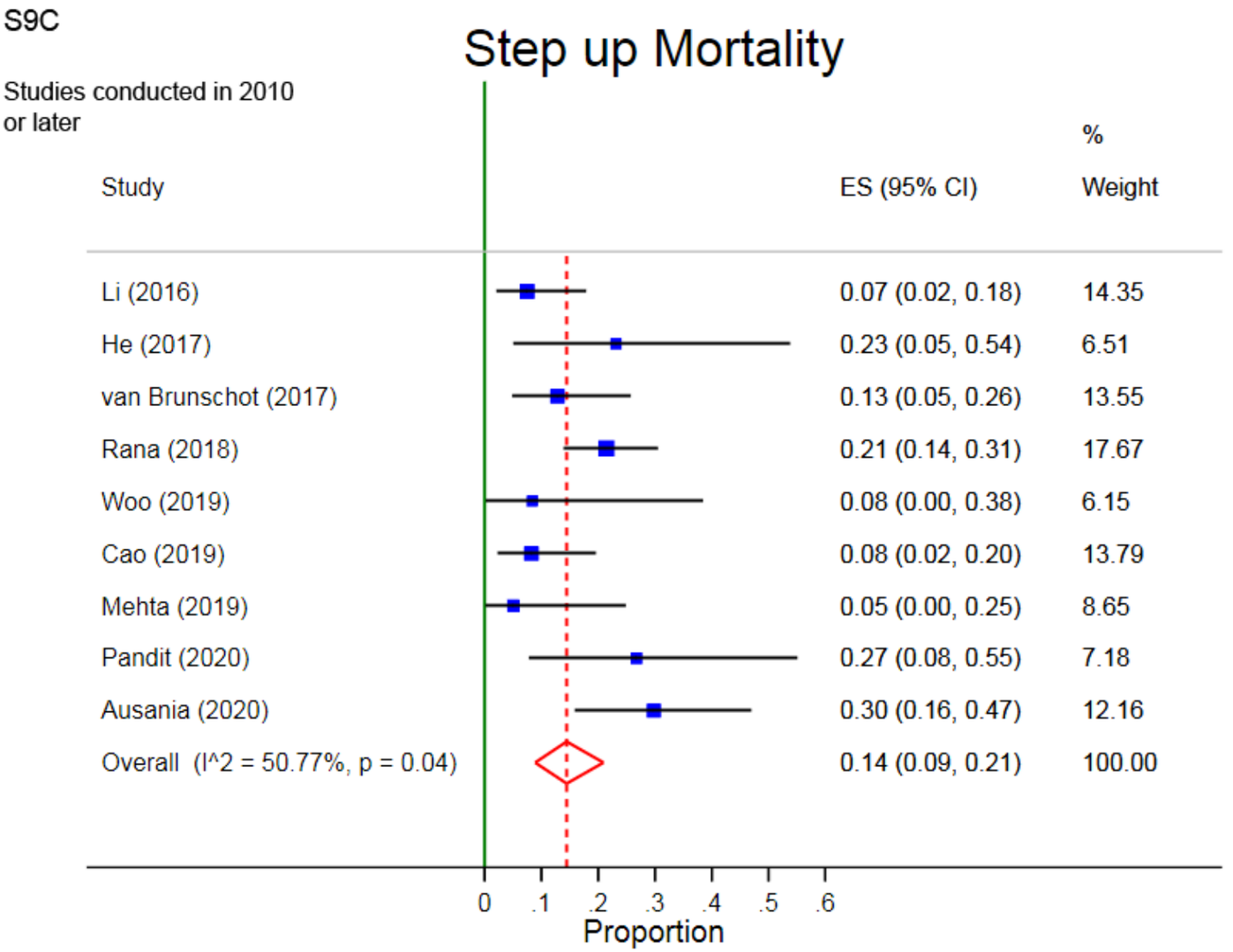
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S9C



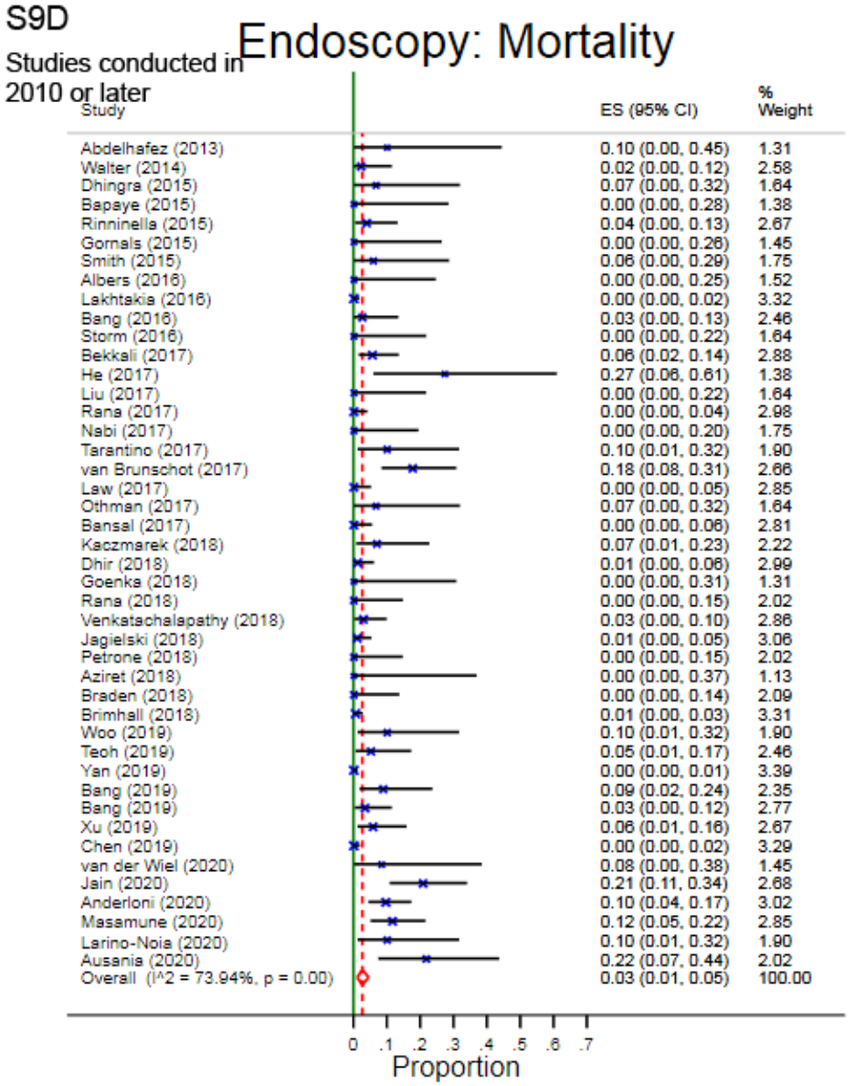
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S9D



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Supplementary material

Table S1. Summary of non-comparative studies

Author (reference number)	Year	Country	Type of study	Type of drainage procedure	Number of patients	Mortality rate (%)
Siddiqui, et al. (1)	2017	USA	Descriptive	Endoscopic	313	0
Yan, et al. (2)	2019	USA	Cohort	Endoscopic	271	0
Bekkali, et al. (3)	2017	UK	Descriptive	Endoscopic	72	5.5
Bapaye, et al. (4)	2015	India+Japan+ Thailand	Cohort	Endoscopic	11	0
Bapaye, et al. (5)	2016	India	Cohort	Endoscopic	133	5.2
Tamura, et al. (6)	2019	Japan	Cohort	Endoscopic	19	0
Smith, et al. (7)	2019	US	Descriptive	Endoscopic	80	3.8
Rana, et al. (8)	2018	India	Descriptive	Endoscopic	86	0
Matsumoto, et al. (9)	2019	Japan	Cohort	Endoscopic	23	4.3
Ahmed, et al. (10)	2017	Ireland	Descriptive	Step up	38	5.3
Maatman, et al. (11)	2019	USA	Descriptive	Open surgery	75	2.7
Teoh, et al. (12)	2020	Hongkong,India, Thailand, S.Korea	Descriptive	Endoscopic	39	5.3
Jagielski, et al. (13)	2018	Poland	Descriptive	Endoscopic	101	1
Jagielski, et al. (14)	2015	Poland	Descriptive	Endoscopic	176	1.1

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Al-Sarireh, et al. (15)	2018	UK	Descriptive	MIS	20	45
Chantarojanasiri, et al. (16)	2018	Japan	Descriptive	Endoscopic	28	7.1
Woo, et al. (17)	2017	Australia	Cohort	Open surgery	10	10
Gardner, et al. (18)	2009	USA	Descriptive	Endoscopic	45	0
Wani, et al. (19)	2011	India	Case series	MIS	15	0
Kaczmarek, et al. (20)	2018	Germany	Case series	Endoscopic	29	6.89
Petrone, et al. (21)	2018	Spain	Descriptive	Endoscopic	23	0
Brimhall, et al. (22)	2018	USA	Descriptive	Endoscopic	197	0.5
Dhingra, et al. (23)	2015	India	Observational	Endoscopic	15	6.7
Mehta, et al. (24)	2019	India	Descriptive	Step up	20	5
Trikudanathan, et al. (25)	2018	USA	Retrospective	Endoscopic	193	7.8
Venkatachalapathy, et al. (26)	2018	Ireland	Descriptive	Endoscopic	70	2.9
Werge, et al. (27)	2018	Denmark	Descriptive	Endoscopic	91	13.2
Schimdt, et al. (28)	2015	Denmark	Descriptive	Endoscopic	81	11
Smith, et al. (29)	2015	USA	Descriptive	Endoscopic	17	5.9
Bang, et al. (30)	2014	USA	Descriptive	Endoscopic	100	4

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Braden, et al. (31)	2018	UK, Germany	Descriptive	Endoscopic	25	0
Othman, et al. (32)	2017	USA	Case series	Endoscopic	15	6.7
Abu Dayye, et al. (33)	2018	USA	Descriptive	Endoscopic	94	3.2
Cao, et al. (34)	2017	China	Cohort	Step up	74	6.8
Bansal, et al. (35)	2017	India	Descriptive	Endoscopic	64	0
Law, et al. (36)	2018	Spain, Hongkong	Descriptive	Endoscopic	68	0
Hyun, et al. (37)	2019	USA	Cohort	Endoscopic	211	2.4
Chen, et al. (38)	2019	USA, Italy, Germany, Canada	Descriptive	Endoscopic	189	0
Dhir, et al. (39)	2018	India	Descriptive	Endoscopic	88	1.1
Tarantino, et al. (40)	2017	Italy	Descriptive	Endoscopic	20	10
Tong, et al. (41)	2017	China	Descriptive	Open surgery	51	35.3
Sileikis, et al. (42)	2017	Lithuania	Descriptive	Open surgery	53	64.2
Watanabe, et al. (43)	2017	Japan	Descriptive	Endoscopic	40	7.5
Bang, et al. (44)*	2019	USA	RCT	Endoscopic	60	3.3
Bang, et al. (45)	2017	USA	Case-control	Endoscopic	39	2.6
Mukai, et al. (46)	2015	Japan	Descriptive	Endoscopic	70	2.9
Xu, et al. (47)	2020	USA	Descriptive	Endoscopic	52	5.8

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Albers, et al. (48)	2016	Germany	Case series	Endoscopic	13	0
Lakhtakia, et al. (49)	2017	India	Descriptive	Endoscopic	205	0
Thompson, et al. (50)	2016	USA	Descriptive	Endoscopic	60	0
Ren, et al. (51)	2017	China	Case series	Endoscopic	18	0
Sharma, et al. (52)	2016	India	Descriptive	Endoscopic	35	2.9
Storm, et al. (53)	2016	USA	Descriptive	Endoscopic	15	0
Rinninella, et al. (54)	2015	Italy, Netherland, Spain, Denmark, Germany, France	Descriptive	Endoscopic	52	3.8
Kitamura, et al. (55)	2016	Japan	Descriptive	Endoscopic	17	0
Cresswell, et al. (56)	2015	UK	Case series	MIS	13	0
Gerin, et al. (57)	2016	France	Descriptive	MIS	11	0
Gornals, et al. (58)	2016	Spain	Case series	Endoscopic	12	0
Madenci, et al. (59)	2014	USA	Descriptive	Open surgery	68	14.7
Gotzinger, et al. (60)	2002	Austria	Cohort	Open surgery	340	39.1
Mathew, et al. (61)	2014	India	Descriptive	MIS	28	3.6
Sarkaria, et al. (62)	2014	USA	Descriptive	Endoscopic	17	0

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Hocke, et al. (63)	2008	Germany	Descriptive	Endoscopic	19	10.5
Voermans, et al. (64)*	2011	Netherlands	RCT	Endoscopic	25	0
Voermans, et al. (65)	2007	Netherlands	Cohort	Endoscopic	25	0
Charnley, et al. (66)	2006	UK	Descriptive	Endoscopic	13	15.4
Walter, et al. (67)	2015	Spain; Germany; Denmark; Netherland	Descriptive	Endoscopic	46	2.2
Worhunsky, et al. (68)	2014	USA	Descriptive	MIS	21	4.8
Zhao, et al. (69)	2015	China	Case series	MIS	17	5.9
Abdelhafez, et al. (70)	2013	Egypt	Case series	Endoscopic	10	10
Gou, et al. (71)	2013	China	Cohort	Open surgery	276	6.2
Reddy, et al. (72)	2006	India	Descriptive	Open surgery	118	38.1
Howard, et al. (73)	2007	USA	Cohort	Open surgery	102	11.8
Bruennler, et al. (74)	2008	Germany	Descriptive	Step up	80	33.8
Rische, et al. (75)	2013	Germany	Descriptive	Endoscopic	31	9.7
Yasuda, et al. (76)	2013	Japan	Descriptive	Endoscopic	57	14
Jurgensen, et al. (77)	2012	Germany	Descriptive	Endoscopic	35	5.7
Rana, et al. (78)	2013	India	Case series	Endoscopic	20	0

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Will, et al. (79)	2012	Germany	Descriptive	Endoscopic	18	5.6
Munene, et al. (80)	2011	Canada	Case series	Open surgery	10	0
Sleeman, et al. (81)	2011	USA	Descriptive	Step up	62	8.1
Tang, et al. (82)	2011	China	Descriptive	MIS	15	0
Cheung, et al. (83)	2010	Hong Kong	Descriptive	Step up	26	26.9
Harrison, et al. (84)	2010	USA	Descriptive	Open surgery	73	13.7
De Rai, et al. (85)	2010	Italy	Descriptive	Open surgery	29	20.7
Sileikis, et al. (86)	2013	Lithuania	Descriptive	MIS	13	0
Dong, et al. (87)	2008	China	Descriptive	Open surgery	53	17
Escourrou, et al. (88)	2008	France	Case series	Endoscopic	12	0
Gui, et al. (89)	2008	Italy	Descriptive	Open surgery	34	26.5
Kingham, et al. (90)	2008	USA	Descriptive	Open surgery	29	13.8
Papachristou, et al. (91)	2007	USA	Descriptive	Endoscopic	53	5.7
Parekh (92)	2006	USA	Descriptive	MIS	18	11.1
Olakowski, et al. (93)	2006	Poland	Descriptive	Open surgery	126	20.6
Funariu, et al. (94)	2006	Romania	Descriptive	Open surgery	80	32.5
Chang, et al. (95)	2006	Taiwan	Case series	Open surgery	19	15.8

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Castellanos, et al. (96)	2013	Spain	Observational	MIS	32	15.6
Tzovaras, et al. (97)	2004	UK	Descriptive	Open surgery	44	18.2
Baron, et al. (98)	2002	USA	Descriptive	Endoscopic	43	4.7
Castellanos, et al. (99)	2002	Spain	Case series	MIS	15	26.7
Ashley, et al. (100)	2001	USA	Descriptive	Open surgery	36	11
Masamune, et al. (101)	2020	Japan	Case series	Endoscopic	69	11.6
Poves, et al. (102)	2000	Spain		Open surgery	49	44.9
Paye, et al. (103)	1999	France	Case series	Open surgery	44	22.7
Tsiotos, et al. (104)	1998	USA	Descriptive	Open surgery	72	25
Takeda, et al. (105)	1998	Japan	Descriptive	Open surgery	179	27.4
Oleynikov, et al. (106)	1998	USA	Descriptive	Open surgery	26	19.2
Fernandez-del Castillo, et al. (107)	1998	USA	Descriptive	Open surgery	64	6.3
Gentile, et al. (108)	1998	USA	Descriptive	Open surgery	40	30
Branum, et al. (109)	1998	USA	Descriptive	Open surgery	50	12
Li, et al. (110)	2016	China	Descriptive	Step up	54	7.4
Jain, et al. (111)	2020	India	Cohort	Endoscopic	53	20.8
Buchler, et al. (112)	2000	Switzerland	Descriptive	Open surgery	28	21

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Bessellink, et al. (113)	2007	Netherlands	Descriptive	Open surgery	53	35.8
Beattie, et al. (114)	2002	UK	Observational	Open surgery	54	42.6
Tsiotos, et al. (115)¶	1998	USA	Descriptive	Open surgery	44	NA
Davies, et al. (116)	2006	UK	Descriptive	Open surgery	18	16.7
Sivasankar, et al. (117)	2006	India	Descriptive	Open surgery	18	27.8
van der Weil, et al. (118)	2020	Germany, Netherland	Case series	Endoscopic	12	8.3
Luckhurst, et al. (119)	2020	USA	Cohort	Open surgery	88	10.2
Bartholdy, et al. (120)	2020	Denmark	Descriptive	Endoscopic	161	16.8
Liu, et al. (121)	2020	China	Descriptive	MIS	164	6.1
Larino-Noia, et al. (122)	2020	Spain	Descriptive	Endoscopic	20	10
Husu, et al. (123)	2020	Finland	Cohort	Open surgery	109	22.9
Pandit, et al. (124)	2020	India	Descriptive	Step up	15	26.7
Carter, et al. (125)	2000	Scotland	Case series	Endoscopic	14	14.3
Moyer, et al. (126)	2019	USA	Case series	Endoscopic	23	8.7
Jurgensen, et al. (127)	2017	Germany	Case series	Endoscopic	14	7.1
Goenka, et al. (128)	2018	India	Descriptive	Endoscopic	10	0
Nabi, et al. (129)	2017	India	Case series	Endoscopic	17	0

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Anderloni, et al. (130)	2020	Italy, USA, Germany	Descriptive	Endoscopic	94	9.6
Wang, et al. (131)	2018	China	Case series	MIS	18	0
Ji, et al. (132)	2018	China	Descriptive	Step up	329	6.4
Chandrasekaran, et al. (133)	2015	India	Cohort	Open surgery	21	NA*
Wig, et al. (134)	2004	India	Descriptive	Open surgery	58	29.3

Abbreviations: MIS: Minimally invasive surgery; RCT: Randomized controlled trial; NA: Not applicable

*This is an RCT study, however endoscopic drainage was used in both arms of the study.

¶Long term outcomes were reported in the survivors, and mortality rate was not applicable

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Table S2. Summary of Comparative Studies Comparing Various Drainage Procedures

Author (reference number)	Year	Country	Type of study	First comparative arm			Second comparative arm			Third comparative arm		
				Procedure	Number	Mortality rate (%)	Procedure	Number	Mortality rate (%)	Procedure	Number	Mortality rate (%)
Woo, et al. (135)	2019	Australia	Descriptive	Endoscopic	20	10	Step up	12	8.3			
Kulkarni, et al. (136)	2015	USA	Descriptive	MIS	8	0	Open surgery	14	0			
Bang, et al. (137)	2019	USA	RCT	Endoscopic	34	8.8	MIS	32	6.3			
Rana, et al. (138)	2018	India	Descriptive	Endoscopic	23	0	Step up	103	21.4			
Dua, et al. (139)	2018	USA	Descriptive	Endoscopic	34	0	MIS	40	2.5			
van Santvoort, et al. (140)	2010	Netherland	RCT	Open surgery	45	16	Step up	43	18.6			
van Brunschot, et al. (141)	2018	8 countries	Cohort	Endoscopic	346	6.6	MIS	467	17.7	Open surgery	198	24.2
Pupelis, et al. (142)	2015	Latvia	Descriptive	MIS	31	6.5	Open surgery	39	12.8			
Jones, et al. (143)	2018	USA	Cohort	Endoscopic	16	6.3	Open surgery	33	21.9			
Aziret, et al. (144)	2018	Turkey	Descriptive	Endoscopic	8	0	Open surgery	13	0			
He, et al. (145)	2017	China	Cohort	Endoscopic	11	27.3	Step up	13	23.1			
Liu, et al. (146)	2017	China	Cohort	Endoscopic	15	0	Open surgery	12	16.7			
van Brunschot, et al. (147)	2018	Netherland	RCT	Endoscopic	51	17.6	Step up	47	12.8			

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Aparna, et al. (148)	2017	India	Descriptive	Open surgery	15	13.3	Step up	9	0			
Gomatos, et al. (149)	2016	UK	Descriptive	MIS	274	15.3	Open surgery	120	23.3			
Keane, et al. (150)	2016	UK	Cohort	Endoscopic	38	0	Step up	30	13.3			
Russell, et al. (151)	2017	New Zealand	Descriptive	Open surgery	13	23.1	MIS	10	20			
Khreiss, et al. (152)	2015	USA	Descriptive	Endoscopic	20	0	MIS	20	0			
Guo, et al. (153)	2016	China	Retrospective	Open surgery	235	19.6	Step up	51	5.9			
Rana, et al. (154)	2014	India	Descriptive	Endoscopic	26	NA*	Open surgery	25	NA*			
Kumar, et al. (155)	2014	USA	Cohort	Endoscopic	12	0	Step up	12	8.3			
Szeliga, et al. (156)	2014	Poland	Descriptive	MIS	12	16.7	Open surgery	12	50	Step up	22	13.6
Pascual, et al. (157)	2013	Spain	Descriptive	Open surgery	21	42.9	Step up	18	16.7			
Bakker, et al. (158)	2012	Netherland	RCT	Endoscopic	10	10	MIS	10	40			
Bausch, et al. (159)	2012	Germany	Descriptive	Open surgery	30	63.3	MIS	32	12.5			
Senthil Kumar, et al. (160)	2012	India	Descriptive	MIS	15	6.7	Open surgery	15	6.7			
Tan, et al. (161)	2012	China	Descriptive	MIS	25	4	Open surgery	51	5.9			
Garg, et al. (162)	2010	India	Observational	Open surgery	18	66.7	Step up	50	28			
van Santvoort, et al. (163)	2007	Netherland	Descriptive	MIS	15	6.7	Open surgery	15	40			
Besselink, et al. (164)	2006	Netherland	Descriptive	MIS	18	11.1	Open surgery	88	38.6			

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Connor, et al. (165)	2005	UK	Descriptive	MIS	47	19.1	Open surgery	41	39			
Gambiez, et al. (166)	1998	France	Descriptive	MIS	20	10	Open surgery	9	33.3			
Tu, et al. (167)	2013	China	Descriptive	MIS	18	5.6	Open surgery	32	12.5			
Cao, et al. (168)	2020	China	Cohort	MIS	45	8.9	Step up	49	8.2			
Garg, et al. (169)	2020	India	RCT	Endoscopic	25	0	MIS	24	0			
Ausania, et al. (170)	2020	Spain	Descriptive	Endoscopic	23	21.7	Open surgery	17	41.2	Step up	37	29.7

Abbreviations: MIS: Minimally invasive surgery; NA: Not applicable

*Long term outcomes were reported in the survivors, and mortality rate was not applicable

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Table S3: Quality assessment of included descriptive and observational studies (STROBE Statement—checklist)

Study, Year (Reference)	Study type	1)a	1)b	2	3	4	5	6)a	6)b	7	8	9	10	11	12)a	12)b	12)c	12)d	12)e	13)a	13)b	13)c	14)a	14)b	14)c	15	16)a	16)b	16)c	17	18	19	20	21	22	Total Score
Siddiqui et al., 2017 (1)	Descriptive	Y	Y	Y	Y	N	Y	Y	N	Y	N	N	N	N	Y	N	N	N	N	Y	N	N	Y	N	N	N	Y	N	NA	N	Y	Y	Y	N	Y	**16**
Woo et al., 2019 (135)	Descriptive	Y	Y	Y	Y	N	Y	Y	N	Y	N	Y	Y	Y	Y	Y	N	NA	N	NA	NA	N	Y	N	NA	NA	Y	N	NA	NA	Y	Y	Y	N	N	**24**
Yan et al., 2019 (2)	Cohort	Y	Y	Y	N	N	Y	N	Y	N	N	Y	N	Y	Y	Y	N	Y	NA	Y	NA	N	Y	Y	N	Y	Y	N	NA	NA	Y	Y	Y	Y	Y	**24**
Kulkarni et al., 2015 (136)	Descriptive	Y	Y	Y	Y	N	Y	Y	Y	Y	N	NA	N	Y	Y	Y	N	NA	Y	Y	NA	N	Y	Y	Y	Y	Y	N	NA	Y	Y	Y	Y	Y	Y	**28**
Bekkali et al., 2017 (3)	Descriptive	Y	Y	Y	Y	N	Y	Y	Y	Y	N	N	N	Y	Y	Y	Y	NA	N	Y	Y	N	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	**28**
Bapaye et al., 2015 (4)	Cohort	Y	Y	Y	Y	N	Y	Y	N	Y	N	N	N	Y	Y	Y	N	N	Y	Y	N	N	Y	Y	Y	Y	Y	N	NA	N	Y	Y	Y	N	N	**21**
Bapaye, et al., 2017 (5)	Cohort	Y	Y	Y	Y	N	Y	Y	Y	Y	N	N	N	Y	Y	Y	Y	NA	Y	Y	N	N	Y	N	Y	Y	Y	N	NA	N	Y	Y	Y	N	N	**23**
Tamura et al., 2019 (6)	Cohort	Y	Y	Y	Y	N	Y	Y	N	Y	N	N	N	Y	Y	N	N	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	N	N	NA	Y	Y	Y	N	Y	**24**
Smith et al., 2019 (7)	Descriptive	Y	Y	Y	Y	N	Y	Y	Y	Y	N	N	N	Y	Y	Y	Y	N	NA	Y	N	Y	Y	Y	Y	Y	Y	Y	NA	NA	Y	N	Y	N	N	**25**
Rana et al., 2018 (138)	Descriptive	Y	Y	Y	Y	N	Y	Y	Y	Y	Y	N	N	Y	N	N	N	NA	Y	Y	Y	N	Y	Y	Y	Y	Y	N	NA	NA	Y	Y	Y	N	Y	**25**

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Rana et al., 2017 (8)	Descriptive	Y	Y	Y	Y	N	Y	Y	Y	Y	N	N	Y	Y	Y	N	N	NA	Y	Y	N	N	Y	N	Y	Y	Y	N	NA	N	Y	Y	Y	N	N	**22**
Matsumoto et al., 2019 (9)	Cohort	N	N	Y	Y	N	N	Y	N	N	N	N	N	N	N	N	N	N	N	N	N	Y	Y	N	N	Y	N	N	NA	N	Y	N	N	N	N	**8**
Ahmed et al., 2017 (10)	Descriptive	Y	Y	Y	Y	N	Y	NA	NA	N	N	NA	NA	Y	N	NA	N	NA	NA	Y	NA	NA	Y	Y	Y	Y	Y	NA	NA	Y	Y	Y	Y	N	N	**27**
Maatman, et al., 2019 (11)	Descriptive	Y	Y	Y	Y	N	Y	Y	N	Y	N	N	N	Y	N	N	Y	NA	N	Y	N	N	Y	Y	N	Y	Y	N	NA	N	N	Y	Y	N	N	**18**
Teoh et al., 2020 (12)	Descriptive	Y	Y	Y	Y	N	Y	N	N	Y	N	N	N	Y	N	N	N	N	Y	Y	N	Y	Y	Y	Y	Y	Y	NA	NA	NA	Y	N	Y	N	N	**20**
Jagielski et al., 2018 (13)	Descriptive	Y	Y	Y	Y	N	Y	N	N	N	N	NA	NA	Y	N	NA	N	NA	NA	Y	NA	NA	Y	Y	Y	Y	Y	NA	NA	NA	Y	Y	Y	NA	N	**26**
Jagielski et al., 2015 (14)	Descriptive	Y	Y	Y	Y	N	Y	N	N	N	N	NA	NA	Y	N	NA	N	NA	NA	Y	NA	NA	Y	Y	Y	Y	Y	NA	NA	NA	Y	Y	Y	NA	N	**26**
Al-Sarireh et al., 2018 (15)	Descriptive	Y	Y	Y	Y	N	Y	Y	N	N	N	NA	NA	Y	N	NA	N	N	NA	Y	NA	NA	Y	Y	Y	Y	Y	NA	NA	NA	Y	N	Y	NA	N	**25**
Chantarojanasiri et al., 2018 (16)	Descriptive	Y	Y	Y	Y	N	Y	Y	Y	Y	Y	N	N	N	N	N	N	NA	Y	Y	N	Y	Y	Y	N	Y	Y	N	N	N	Y	Y	Y	N	N	**5**
Woo et al., 2017 (17)	cohort	Y	Y	Y	Y	N	Y	Y	Y	Y	N	N	N	Y	N	N	N	Y	Y	Y	N	N	Y	Y	N	Y	Y	N	N	NA	Y	Y	Y	N	N	**20**
Dua, et al., 2018 (139)	Descriptive	Y	Y	Y	Y	N	Y	Y	Y	Y	Y	N	N	N	Y	N	N	NA	Y	Y	N	Y	Y	Y	N	Y	Y	N	N	N	Y	Y	Y	N	Y	**22**

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van Brunschot et al., 2018 (141)	Cohort	Y	Y	Y	Y	N	Y	Y	Y	Y	Y	N	Y	N	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	NA	NA	NA	Y	N	Y	N	Y	**29**
Gardner et al., 2009 (18)	Descriptive	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	N	N	Y	Y	Y	Y	NA	Y	Y	N	N	Y	Y	Y	Y	Y	N	N	Y	Y	Y	Y	Y	N	**27**
Wani, et al., 2011 (19)	Case series	Y	Y	Y	Y	N	Y	NA	NA	N	N	NA	NA	Y	N	NA	N	NA	NA	Y	NA	NA	Y	Y	Y	Y	Y	NA	NA	NA	Y	N	Y	NA	Y	**28**
Kaczmarek et al., 2018 (20)	Case series	Y	Y	Y	Y	N	Y	NA	NA	N	N	NA	NA	Y	N	NA	N	NA	NA	Y	NA	NA	Y	Y	Y	Y	Y	NA	NA	NA	Y	N	Y	NA	N	**27**
Petrone et al., 2018 (21)	Descriptive	Y	Y	Y	Y	Y	Y	Y	Y	Y	N	N	N	Y	N	Y	Y	NA	Y	Y	N	N	Y	Y	N	Y	Y	N	Y	Y	Y	N	Y	N	Y	**24**
Brimhall et al., 2018 (22)	Descriptive	Y	Y	Y	Y	N	Y	Y	Y	Y	N	N	N	Y	Y	Y	Y	Y	Y	Y	N	Y	Y	Y	N	Y	Y	N	NA	NA	Y	Y	Y	N	N	**25**
Dhingra et al., 2015 (23)	Observational	Y	Y	Y	Y	N	Y	NA	NA	Y	N	N	N	Y	NA	NA	N	NA	NA	NA	NA	Y	Y	Y	Y	Y	NA	NA	NA	Y	Y	Y	Y	N	N	**27**
Pupelis et al., 2015 (142)	Prospective analysis (descriptive)	Y	Y	Y	N	Y	Y	NA	NA	Y	N	N	N	Y	N	N	N	N	NA	NA	N	NA	Y	Y	NA	Y	NA	NA	NA	N	Y	N	Y	N	N	**27**
Mehta et al., 2019 (24)	Descriptive	Y	Y	Y	Y	N	Y	NA	NA	Y	N	N	N	Y	NA	NA	N	N	N	N	Y	Y	Y	Y	N	Y	Y	N	Na	NA	Y	N	Y	N	Y	**22**
Trikudanathan et al., 2018 (25)	Retrospective analysis	Y	Y	Y	Y	N	Y	Y	Y	Y	N	N	N	Y	N	N	N	NA	Y	Y	Y	Y	Y	Y	N	Y	Y	N	NA	N	Y	Y	Y	N	Y	**23**
Venkatachalapathy et al.,	Descriptive	Y	Y	Y	Y	N	Y	NA	N	Y	N	N	N	N	N	N	N	N	N	Y	N	N	Y	Y	N	Y	Y	Y	NA	Y	Y	Y	Y	N	N	**18**

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2018 (26)																																				
Werge et al., 2018 (27)	Descriptive	Y	Y	Y	Y	N	Y	N	N	Y	N	N	N	Y	N	N	N	N	NA	Y	N	N	Y	Y	N	Y	Y	N	NA	NA	Y	Y	Y	N	Y	**19**
Schmidt et al., 2015 (28)	Descriptive	Y	Y	Y	Y	N	Y	N	N	Y	N	N	N	N	N	N	N	N	NA	Y	N	N	Y	Y	N	Y	Y	N	N	NA	Y	Y	Y	N	N	**16**
Smith et al., 2015 (29)	cohort	Y	Y	Y	Y	N	Y	Y	N	N	N	N	N	Y	N	N	N	Y	N	Y	N	N	Y	Y	N	Y	Y	N	NA	NA	Y	Y	Y	N	N	**18**
Bang et al., 2014 (30)	Descriptive	Y	Y	Y	Y	N	Y	Y	N	Y	Y	N	N	Y	Y	N	N	N	N	Y	N	N	Y	Y	Y	Y	Y	Y	NA	NA	Y	Y	Y	N	N	**22**
Braden et al., 2018 (31)	Descriptive	Y	Y	Y	Y	Y	Y	N	N	Y	Y	N	N	Y	N	N	N	N	NA	Y	N	N	Y	Y	N	Y	Y	NA	NA	NA	Y	Y	Y	N	N	**21**
Jones et al., 2018 (143)	cohort	Y	Y	Y	Y	N	Y	Y	N	Y	Y	N	N	Y	N	N	N	N	NA	Y	N	N	Y	Y	N	Y	Y	Y	NA	NA	Y	N	Y	N	N	**20**
Othman et al., 2017 (32)	Case series	Y	Y	Y	Y	N	Y	NA	NA	N	N	NA	NA	Y	N	NA	N	NA	NA	Y	NA	NA	Y	Y	Y	Y	Y	NA	NA	NA	Y	N	Y	NA	N	**27**
Abu Dayye et al., 2018 (33)	Descriptive	Y	Y	Y	Y	N	Y	Y	Y	Y	N	N	N	Y	N	N	N	NA	Y	Y	Y	Y	Y	Y	N	Y	Y	N	NA	N	Y	Y	Y	N	Y	**23**
Aziret et al., 2018 (144)	Descriptive	Y	Y	Y	Y	N	Y	Y	N	Y	Y	N	N	Y	Y	N	N	N	Y	Y	N	NA	Y	Y	Y	Y	Y	Y	NA	NA	Y	Y	Y	Y	N	**25**
Cao et al., 2017 (34)	Cohort	Y	Y	Y	Y	N	Y	Y	Y	Y	Y	N	Y	Y	Y	Y	N	N	Y	Y	N	Y	Y	Y	Y	Y	Y	Y	NA	NA	Y	Y	Y	N	N	**27**
He et al., 2017 (145)	Cohort	Y	Y	Y	Y	N	Y	Y	Y	Y	Y	N	Y	Y	Y	Y	N	N	Y	Y	N	Y	Y	Y	Y	Y	Y	Y	NA	NA	Y	Y	Y	N	N	**27**

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Bansal et al., 2017 (35)	Descriptive	Y	Y	Y	Y	N	Y	NA	NA	Y	NA	NA	N	Y	N	NA	N	N	NA	Y	N	Y	Y	Y	Y	Y	Y	Y	NA	NA	Y	Y	Y	NA	N	**27**
Law et al., 2018 (36)	Descriptive	Y	Y	Y	Y	N	Y	Y	N	Y	Y	N	N	Y	Y	N	N	N	Y	Y	N	NA	Y	Y	Y	Y	Y	Y	NA	NA	Y	Y	Y	Y	N	**25**
Liu et al., 2017 (146)	cohort	Y	Y	Y	Y	N	Y	Y	Y	Y	Y	N	N	Y	Y	Y	N	Y	Y	Y	N	Y	Y	Y	Y	Y	Y	Y	NA	NA	Y	N	Y	Y	N	**27**
Hyun et al., 2019 (37)	cohort	Y	Y	Y	Y	N	Y	Y	Y	Y	Y	N	N	Y	Y	Y	N	Y	Y	Y	N	Y	Y	Y	Y	Y	Y	Y	NA	NA	Y	Y	Y	N	N	**27**
Chen et al., 2019 (38)	Descriptive	Y	Y	Y	Y	N	Y	Y	N	Y	Y	N	N	Y	Y	N	N	N	NA	Y	N	N	Y	Y	Y	Y	Y	Y	NA	NA	Y	Y	Y	Y	N	**24**
Dhir et al., 2018 (39)	Descriptive	Y	Y	Y	Y	N	Y	Y	Y	Y	N	N	N	Y	Y	Y	Y	N	NA	Y	N	Y	Y	Y	Y	Y	Y	Y	NA	NA	Y	N	Y	N	N	**25**
Tarantino et al., 2017 (40)	Descriptive	Y	Y	Y	Y	N	Y	NA	NA	N	N	NA	NA		N	NA	N	NA	NA	Y	NA	NA	Y	Y	Y	Y	Y	NA	NA	NA	Y	N	Y	NA	N	**27**
Tong et al., 2017 (41)	Descriptive	Y	Y	Y	Y	N	Y	Y	Y	Y	N	Y	N	Y	N	Y	N	Y	N	Y	NA	Y	Y	Y	Y	Y	Y	Y	NA	NA	Y	N	Y	N	N	**25**
Sileikis et al., 2017 (42)	Descriptive	Y	Y	Y	Y	N	Y	NA	NA	N	N	NA	NA	Y	N	Y	N	N	Y	Y	N	N	Y	Y	N	Y	Y	Y	NA	NA	Y	N	Y	N	N	**22**
Watanabe et al., 2017 (43)	Descriptive	Y	Y	Y	Y	N	Y	Y	N	N	N	NA	NA	Y	N	Y	N	N	Y	Y	N	N	Y	Y	N	Y	Y	Y	NA	NA	Y	N	Y	N	N	**21**
Bang et al., 2017 (45)	case–control	Y	Y	Y	Y	N	Y	Y	Y	Y	Y	N	N	Y	Y	N	Y	N	Y	Y	N	N	Y	Y	NA	Y	Y	Y	Y	Y	Y	Y	Y	N	N	**25**
Mukai et al., 2015 (46)	Descriptive	Y	Y	Y	Y	N	Y	NA	NA	N	N	NA	NA	Y	N	NA	N	NA	NA	Y	NA	NA	Y	Y	Y	Y	Y	NA	NA	NA	Y	Y	Y	NA	N	**28**

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Aparna et al., 2017 (148)	Descriptive	Y	Y	Y	Y	N	Y	NA	NA	N	N	NA	NA	Y	N	NA	N	NA	NA	Y	NA	N	Y	Y	Y	Y	Y	NA	NA	NA	Y	N	Y	NA	N	**26**
Xu al., 2020 (47)	Descriptive	Y	Y	Y	Y	N	Y	N	N	Y	N	N	N	Y	N	N	N	N	Y	Y	N	Y	Y	Y	Y	Y	Y	NA	NA	NA	Y	N	Y	N	N	**20**
Albers et al., 2016 (48)	case series	Y	Y	Y	Y	N	Y	N	N	Y	N	N	N	Y	N	N	N	N	Y	Y	N	NA	Y	Y	Y	Y	Y	NA	NA	NA	Y	Y	Y	N	N	**21**
Gomatos et al., 2016 (149)	Descriptive	Y	Y	Y	Y	N	Y	NA	NA	N	N	N	N	Y	N	N	N	NA	NA	Y	N	N	Y	Y	Y	Y	Y	NA	NA	NA	Y	Y	Y	N	N	**22**
Keane et al., 2016 (150)	cohort	Y	Y	Y	Y	N	Y	Y	Y	Y	Y	N	N	Y	Y	Y	N	Y	Y	Y	N	Y	Y	Y	Y	Y	Y	Y	NA	NA	Y	Y	Y	N	Y	**28**
Lakhtakia et al., 2017 (49)	Descriptive	Y	Y	Y	Y	N	Y	Y	N	N	N	NA	NA	Y	N	NA	N	NA	NA	Y	NA	Y	Y	Y	Y	Y	Y	NA	NA	NA	Y	Y	Y	N	N	**25**
Thompson et al., 2016 (50)	Descriptive	Y	Y	Y	Y	N	Y	NA	NA	N	N	NA	NA	Y	N	NA	N	NA	NA	Y	NA	N	Y	Y	Y	Y	Y	NA	NA	Y	Y	Y	Y	N	Y	**27**
Ren et al., 2017 (51)	case series	Y	Y	Y	Y	N	Y	N	N	Y	N	N	N	Y	N	N	N	N	Y	Y	N	NA	Y	Y	Y	Y	Y	NA	NA	NA	Y	Y	Y	NA	Y	**23**
Russell et al., 2017 (151)	Descriptive	Y	Y	Y	Y	N	Y	NA	NA	N	N	NA	NA	Y	N	NA	N	NA	NA	Y	NA	Y	Y	Y	Y	Y	Y	NA	NA	Y	Y	Y	Y	N	Y	**28**
Sharma et al., 2016 (52)	Descriptive	Y	Y	Y	Y	N	Y	NA	NA	N	N	NA	NA	Y	N	NA	N	NA	NA	Y	NA	Y	Y	Y	Y	Y	Y	NA	NA	Y	Y	Y	Y	N	N	**27**
Storm et al., 2016 (53)	Descriptive	Y	Y	Y	Y	N	Y	NA	NA	N	N	NA	NA	Y	N	NA	N	NA	NA	Y	NA	Y	Y	Y	Y	Y	Y	NA	NA	Y	Y	Y	Y	NA	N	**28**

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Rinninella et al., 2015 (54)	Descriptive	Y	Y	Y	Y	N	Y	NA	NA	N	N	NA	NA	Y	N	NA	N	NA	NA	Y	N	N	Y	Y	Y	Y	Y	NA	NA	Y	Y	Y	Y	N	N	**26**
Khreiss et al., 2015 (152)	Descriptive	Y	Y	Y	Y	N	Y	N	N	N	N	NA	NA	Y	N	NA	N	NA	NA	Y	N	N	Y	Y	Y	Y	Y	NA	NA	Y	Y	Y	Y	N	N	**23**
Kitamura et al., 2016 (55)	Descriptive	Y	Y	Y	Y	N	Y	N	N	N	N	N	NA	Y	N	N	N	NA	NA	Y	N	N	Y	Y	Y	Y	Y	NA	NA	Y	Y	Y	Y	N	N	**21**
Cresswell et al., 2015 (56)	case series	Y	Y	Y	Y	N	Y	NA	NA	N	N	NA	NA	Y	N	NA	N	NA	NA	Y	NA		Y	Y	Y	Y	Y	NA	NA	Y	Y	Y	Y	NA	N	**28**
Gerin et al., 2016 (57)	Descriptive	Y	Y	Y	Y	N	Y	N	N	Y	N	N	N	Y	N	N	N	N	Y	Y	N	N	Y	Y	Y	Y	Y	NA	NA	NA	Y	N	Y	N	N	**19**
Gornals et al., 2016 (58)	case series	Y	Y	Y	Y	N	Y	NA	NA	N	N	NA	NA	Y	N	NA	N	NA	NA	Y	NA	Y	Y	Y	Y	Y	Y	NA	NA	Y	Y	Y	Y	NA	N	**28**
Guo et al., 2016 (153)	Retrospective analysis	Y	Y	Y	Y	N	Y	NA	NA	N	N	NA	NA	Y	N	NA	N	NA	NA	Y	NA	Y	Y	Y	Y	Y	Y	NA	NA	Y	Y	Y	Y	N	N	**28**
Rana et al., 2014 (154)	Descriptive	Y	Y	Y	Y	Y	Y	N	N	N	N	NA	NA	Y	N	N	N	N	NA	Y	N	N	Y	Y	Y	Y	Y	NA	NA	Y	Y	Y	Y	N	N	**22**
Madenci et al., 2014 (59)	Descriptive	Y	Y	Y	Y	N	Y	NA	NA	N	N	NA	NA	Y	N	N	N	N	NA	Y	NA	Y	Y	Y	Y	Y	Y	NA	NA	Y	Y	Y	Y	Y	Y	**27**
Gotzinger, et al., 2002 (60)	cohort	Y	Y	Y	Y	N	Y	Y	Y	Y	Y	N	N	Y	Y	Y	N	Y	Y	Y	N	Y	Y	Y	Y	Y	Y	Y	NA	NA	Y	Y	Y	N	N	**27**
Kumar et al., 2014 (155)	cohort	Y	Y	Y	Y	N	Y	Y	Y	Y	Y	N	N	Y	N	N	N	N	Y	Y	N	N	Y	Y	Y	Y	Y	Y	NA	NA	Y	Y	Y	N	N	**23**
Mathew et al.,	Descriptive	Y	Y	Y	Y	N	Y	NA	NA	N	N	NA	NA	Y	N	NA	N	NA	NA	Y	NA	Y	Y	Y	Y	Y	Y	NA	NA	Y	Y	Y	Y	NA	N	**28**

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2014 (61)																																			
Sarkaria et al., 2014 (62)	Descriptive	Y	Y	Y	Y	N	Y	NA	NA	N	N	NA	NA	Y	N	NA	N	NA	NA	Y	NA	NA	Y	Y	Y	Y	Y	NA	NA	Y	Y	Y	Y	N	N	**27**
Hocke et al., 2008 (63)	Descriptive	Y	Y	Y	Y	Y	Y	N	N	N	N	NA	NA	Y	N	N	N	N	NA	Y	N	N	Y	Y	Y	Y	Y	NA	NA	Y	Y	Y	Y	N	N	**22**
Voermans et al., 2011 (65)	cohort	Y	Y	Y	Y	N	Y	Y	Y	Y	Y	N	N	Y	Y	N	N	N	Y	Y	N	N	Y	Y	Y	Y	Y	Y	NA	NA	Y	Y	Y	N	N	**24**
Charnley et al., 2006 (66)	Descriptive	Y	Y	Y	Y	N	Y	N	N	N	N	NA	NA	Y	N	N	N	N	NA	Y	N	N	Y	Y	Y	Y	Y	NA	NA	Y	Y	Y	Y	N	N	**21**
Walter et al., 2015 (67)	Descriptive	Y	Y	Y	Y	N	Y	N	N	Y	N	N	N	Y	N	N	N	N	Y	Y	N	N	Y	Y	Y	Y	Y	NA	NA	NA	Y	N	Y	N	N	**19**
Szeliga et al., 2014 (156)	Descriptive	Y	Y	Y	Y	N	Y	NA	NA	N	N	NA	NA	Y	N	NA	N	NA	NA	Y	NA	Y	Y	Y	Y	Y	Y	NA	NA	Y	Y	N	Y	N	N	**26**
Worhunsky et al., 2014 (68)	Descriptive	Y	Y	Y	Y	N	Y	N	N	Y	N	N	N	Y	N	N	N	N	Y	Y	N	N	Y	Y	N	Y	Y	NA	NA	NA	Y	N	Y	N	N	**18**
Zhao et al., 2015 (69)	case series	Y	Y	Y	Y	N	Y	NA	NA	N	N	NA	NA	Y	N	NA	N	NA	NA	Y	NA	Y	Y	Y	Y	Y	Y	NA	NA	Y	Y	Y	Y	NA	N	**28**
Abdelhafez et al., 2013 (70)	case series	Y	Y	Y	Y	N	Y	NA	NA	N	N	NA	NA	Y	N	NA	N	NA	NA	Y	NA	Y	Y	Y	N	Y	Y	NA	NA	Y	Y	Y	Y	N	N	**26**
Gou et al., 2013 (71)	cohort	Y	Y	Y	Y	N	Y	Y	Y	Y	Y	N	N	Y	N	N	N	N	Y	Y	N	N	Y	Y	N	Y	Y	Y	NA	NA	Y	N	Y	N	N	**21**
Reddy et al.,	Descriptive	Y	Y	Y	Y	N	Y	N	N	Y	N	N	N	Y	N	N	N	N	Y	Y	N	N	Y	Y	N	Y	Y	NA	NA	NA	Y	N	Y	N	N	**18**

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2006 (72)																																			
Howard et al., 2007 (73)	cohort	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	N	N	Y	N	N	N	N	Y	Y	N	N	Y	Y	N	Y	Y	Y	NA	NA	Y	N	Y	N	N	**22**
Bruennler et al., 2008 (74)	Descriptive	Y	Y	Y	Y	N	Y	NA	NA	N	N	NA	NA	Y	N	NA	N	NA	NA	Y	NA	Y	Y	Y	N	Y	Y	NA	NA	Y	Y	N	Y	N	N	**25**
Pascual et al., 2013 (157)	Descriptive	Y	Y	Y	Y	N	Y	N	N	Y	N	N	N	Y	N	N	N	N	Y	Y	N	Y	Y	Y	N	Y	Y	NA	NA	NA	Y	Y	Y	N	N	**20**
Rische al., 2013 (75)	Descriptive	Y	Y	Y	Y	N	Y	Y	N	Y	Y	N	N	Y	N	N	Y	N	NA	Y	N	Y	Y	Y	N	Y	Y	NA	NA	NA	Y	N	Y	N	N	**22**
Yasuda et al., 2013 (76)	Descriptive	Y	Y	Y	Y	N	Y	Y	NA	NA	N	N	NA	NA	Y	N	N	N	NA	Y	Y	Y	Y	Y	N	Y	Y	Y	NA	NA	Y	Y	Y	N	N	**25**
Bausch et al., 2012 (159)	Descriptive	Y	Y	Y	Y	N	Y	Y	NA	NA	N	N	NA	NA	Y	N	N	N	NA	Y	Y	Y	Y	Y	N	Y	Y	Y	NA	NA	Y	Y	Y	N	N	**25**
Jurgensen et al., 2012 (77)	Descriptive	Y	Y	Y	Y	N	Y	Y	N	N	N	N	NA	NA	Y	N	N	N	NA	Y	Y	N	Y	Y	N	Y	Y	Y	NA	NA	Y	Y	Y	N	N	**22**
Rana et al., 2013 (78)	case series	Y	Y	Y	Y	N	Y	NA	NA	N	N	NA	NA	Y	N	NA	N	NA	NA	Y	NA	Y	Y	Y	N	Y	Y	Y	NA	NA	Y	Y	Y	NA	N	**27**
Senthil Kumar et al., 2012 (160)	Descriptive	Y	Y	Y	Y	N	Y	N	N	Y	N	N	N	Y	N	N	N	N	Y	Y	N	N	Y	Y	N	Y	Y	NA	NA	NA	Y	N	Y	N	Y	**19**
Tan et al., 2012 (161)	Descriptive	Y	Y	Y	Y	N	Y	Y	N	N	N	N	NA	NA	Y	N	N	N	NA	Y	Y	Y	Y	Y	N	Y	Y	Y	NA	NA	Y	Y	Y	N	N	**23**

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Will et al., 2012 (79)	Descriptive	Y	Y	Y	Y	N	Y	Y	N	N	N	N	N	N	Y	N	N	N	NA	Y	Y	NA	Y	Y	N	Y	Y	Y	NA	NA	Y	Y	Y	N	N	**21**
Munene et al., 2011 (80)	case series	Y	Y	Y	Y	N	Y	NA	NA	N	N	NA	NA	Y	N	NA	N	NA	NA	Y	Y	NA	Y	Y	Y	Y	Y	Y	NA	NA	Y	N	Y	NA	N	**27**
Sleeman et al., 2011 (81)	Descriptive	Y	Y	Y	Y	N	Y	Y	N	Y	Y	N	N	Y	N	N	Y	N	NA	Y	N	N	Y	Y	N	Y	Y	Y	NA	NA	Y	N	Y	N	N	**21**
Tang et al., 2011 (82)	Descriptive	Y	Y	Y	Y	N	Y	NA	NA	N	N	NA	NA	Y	N	NA	N	NA	NA	Y	Y	NA	Y	Y	Y	Y	Y	Y	NA	NA	Y	N	Y	NA	N	**27**
Cheung et al., 2010 (83)	Descriptive	Y	Y	Y	Y	N	Y	Y	N	Y	Y	N	N	Y	N	N	Y	N	NA	Y	N	Y	Y	Y	N	Y	Y	NA	NA	NA	Y	N	Y	N	N	**22**
Garg et al., 2010 (162)	observational	Y	Y	Y	Y	N	Y	Y	N	Y	Y	N	N	Y	N	N	Y	N	NA	Y	N	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	N	**25**
Harrison et al., 2010 (84)	Descriptive	Y	Y	Y	Y	N	Y	Y	N	Y	Y	N	N	Y	N	N	Y	N	NA	Y	N	N	Y	Y	N	Y	Y	NA	NA	NA	Y	N	Y	N	N	**21**
De Rai et al., 2010 (85)	Descriptive	Y	Y	Y	Y	N	Y	Y	Y	Y	Y	N	N	Y	Y	N	N	N	Y	Y	N	N	Y	Y	N	Y	Y	NA	NA	NA	Y	N	Y	N	N	**22**
Sileikis et al., 2013 (86)	Descriptive	Y	Y	Y	Y	N	Y	NA	NA	N	N	NA	NA	Y	N	NA	N	NA	NA	Y	Y	NA	Y	Y	Y	Y	Y	Y	NA	NA	Y	N	Y	NA	N	**27**
Dong et al., 2008 (87)	Descriptive	Y	Y	Y	Y	N	Y	Y	N	Y	Y	N	N	Y	N	N	Y	N	NA	Y	N	N	Y	Y	N	Y	Y	Y	NA	NA	Y	N	Y	N	N	**21**
Escourrou et al., 2008 (88)	case series	Y	Y	Y	Y	N	Y	NA	NA	N	N	NA	NA	Y	N	NA	N	NA	NA	Y	Y	NA	Y	Y	Y	Y	Y	Y	NA	NA	Y	N	Y	NA	N	**27**

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Gui et al., 2008 (89)	Descriptive	Y	Y	Y	Y	N	Y	Y	N	N	N	N	NA	NA	Y	N	N	N	NA	Y	Y	Y	Y	Y	N	Y	Y	Y	NA	NA	Y	Y	Y	N	N	**23**
Kingham et al., 2008 (90)	Descriptive	Y	Y	Y	Y	N	Y	Y	N	Y	Y	N	N	Y	N	N	Y	N	NA	Y	N	Y	Y	Y	N	Y	Y	NA	NA	NA	Y	N	Y	N	N	**22**
van Santvoort et al., 2007 (163)	Descriptive	Y	Y	Y	Y	N	Y	Y	N	Y	Y	N	N	Y	Y	N	Y	N	NA	Y	N	N	Y	Y	N	Y	Y	NA	NA	NA	Y	N	Y	N	N	**22**
Papachristou et al., 2007 (91)	Descriptive	Y	Y	Y	Y	N	Y	Y	N	Y	Y	N	N	Y	Y	N	N	N	NA	Y	N	N	Y	Y	N	Y	Y	NA	NA	NA	Y	N	Y	N	N	**21**
Besselink et al., 2006 (164)	Descriptive	Y	Y	Y	Y	N	Y	Y	N	Y	Y	N	N	Y	Y	N	N	N	NA	Y	N	N	Y	Y	N	Y	Y	NA	NA	NA	Y	N	Y	N	Y	**22**
Parekh 2006 (92)	Descriptive	Y	Y	Y	Y	N	Y	Y	N	Y	Y	N	N		N	N	Y	N	NA	Y	N	Y	Y	Y	N	Y	Y	NA	NA	NA	Y	N	Y	N	N	**22**
Olakowski et al., 2006 (93)	Descriptive	Y	Y	Y	Y	N	Y	Y	N	Y	Y	N	N	Y	N	N	N	N	NA	Y	N	Y	Y	Y	N	Y	Y	NA	NA	NA	Y	N	Y	N	N	**21**
Funariu et al., 2006 (94)	Descriptive	Y	Y	Y	Y	N	Y	Y	N	Y	Y	N	N	Y	N	N	N	N	NA	Y	N	Y	Y	Y	N	Y	Y	NA	NA	NA	Y	N	Y	N	N	**21**
Chang et al., 2006 (95)	case series	Y	Y	Y	Y	N	Y	NA	NA	N	N	NA	NA	Y	N	NA	N	NA	NA	Y	Y	NA	Y	Y	Y	Y	Y	Y	NA	NA	Y	N	Y	NA	N	**27**
Castellanos et al., 2013 (96)	observational	Y	Y	Y	Y	N	Y	NA	NA	Y	Y	NA	NA	Y	Y	NA	NA	N	NA	Y	Y	Y	Y	Y	Y	Y	Y	Y	NA	NA	Y	N	Y	NA	NA	**31**
Connor et al., 2005 (165)	Descriptive	Y	Y	Y	Y	N	Y	N	N	Y	N	N	N	Y	N	N	N	N	Y	Y	N	N	Y	Y	N	Y	Y	NA	NA	NA	Y	N	Y	N	N	**18**

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Tzovaras et al., 2004 (97)	Descriptive	Y	Y	Y	Y	N	Y	Y	N	Y	Y	N	N	Y	N	N	N	N	NA	Y	N	Y	Y	Y	N	Y	Y	NA	NA	NA	Y	N	Y	N	N	**21**
Baron et al., 2002 (98)	Descriptive	Y	Y	Y	Y	N	Y	Y	N	Y	Y	N	N	Y	Y	NA	N	N	NA	Y	N	Y	Y	Y	N	Y	Y	NA	NA	NA	Y	N	Y	N	N	**23**
Castellanos et al., 2002 (99)	case series	Y	Y	Y	Y	N	Y	NA	NA	N	N	NA	NA	Y	N	NA	N	NA	NA	Y	Y	NA	Y	Y	Y	Y	Y	Y	NA	NA	Y	N	Y	NA	N	**27**
Ashley et al., 2001 (100)	Descriptive	Y	Y	Y	Y	N	Y	Y	N	Y	Y	N	N	Y	Y	NA	N	N	NA	Y	N	Y	Y	Y	N	Y	Y	NA	NA	NA	Y	N	Y	N	N	**23**
Masamune et al., 2020 (101)	case series	Y	Y	Y	Y	N	Y	NA	NA	N	N	NA	NA	Y	N	NA	N	NA	NA	Y	Y	NA	Y	Y	Y	Y	Y	Y	NA	NA	Y	N	Y	NA	N	**27**
Poves et al., 2000 (102)	Descriptive	Y	Y	Y	N	Y	Y	Y	NA	N	N	N	N	N	Y	NA	N	NA	NA	Y	N	NA	Y	Y	NA	Y	N	Y	NA	NA	Y	N	Y	N	N	**22**
Paye et al., 1999 (103)	Case series	Y	Y	Y	Y	N	Y	NA	NA	N	N	NA	NA	Y	N	NA	N	NA	NA	Y	Y	NA	Y	Y	Y	Y	Y	Y	NA	NA	Y	N	Y	NA	N	**27**
Tsiotos, et al., 1998 (104)	Descriptive	Y	Y	Y	Y	N	Y	Y	N	Y	N	N	N	Y	Y	N	N	N	Y	Y	N	N	Y	Y	N	Y	Y	NA	NA	NA	Y	N	N	N	N	**20**
Takeda et al., 1998 (105)	Descriptive	Y	Y	Y	Y	N	Y	Y	Y	Y	N	N	N	Y	Y	Y	N	N	NA	N	N	Y	Y	Y	N	Y	Y	NA	NA	NA	Y	N	Y	N	N	**22**
Oleynikov et al., 1998 (106)	Descriptive	Y	Y	Y	Y	N	Y	Y	N	Y	Y	N	N	Y	Y	NA	N	N	NA	Y	N	Y	Y	Y	N	Y	Y	NA	NA	NA	Y	N	Y	N	N	**23**
Fernandez-del Castillo et al., 1998 (107)	Descriptive	Y	Y	Y	Y	N	Y	Y	N	Y	Y	N	N	Y	Y	N	N	N	NA	Y	N	N	Y	Y	N	Y	Y	NA	NA	NA	Y	N	Y	N	N	**21**

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Gentile et al., 1998 (108)	Descriptive	Y	Y	Y	Y	N	Y	Y	N	Y	Y	N	N	Y	Y	N	N	N	NA	Y	N	N	Y	Y	N	Y	Y	NA	NA	NA	Y	N	Y	N	N	**20**
Gambiez et al., 1998 (166)	Descriptive	Y	Y	Y	Y	N	Y	Y	N	Y	Y	N	N	Y	Y	N	N	N	N	Y	N	Y	Y	Y	N	Y	Y	NA	NA	NA	Y	N	Y	N	N	**21**
Branum et al., 1998 (109)	Descriptive	Y	Y	Y	Y	N	Y	Y	N	Y	Y	N	N	Y	Y	N	N	N	N	Y	N	Y	Y	Y	N	Y	Y	NA	NA	NA	Y	N	Y	N	N	**21**
Li et al., 2016 (110)	Descriptive	Y	Y	Y	Y	N	Y	Y	N	Y	Y	N	N	Y	Y	N	N	N	NA	Y	N	Y	Y	Y	N	Y	Y	NA	NA	NA	Y	N	Y	N	N	**22**
Jain et al., 2020 (111)	cohort	Y	Y	Y	Y	N	Y	Y	Y	Y	Y	N	N	Y	Y	Y	Y	N	NA	Y	N	Y	Y	Y	N	Y	Y	NA	NA	NA	Y	Y	Y	N	N	**26**
Buchler et al., 2000 (112)	Descriptive	Y	Y	Y	Y	N	Y	Y	N	Y	Y	N	N	Y	Y	N	N	N	NA	Y	N	Y	Y	Y	N	Y	Y	NA	NA	NA	Y	N	Y	N	N	**22**
Besselink et al., 2007 (113)	Descriptive	Y	Y	Y	Y	N	Y	Y	N	Y	Y	N	N	Y	Y	Y	Y	N	N	Y	N	Y	Y	Y	N	Y	Y	NA	NA	NA	Y	N	Y	N	Y	**24**
Tu et al., 2013 (167)	Descriptive	Y	Y	Y	Y	N	Y	NA	NA	N	N	NA	NA	Y	N	NA	N	NA	NA	Y	Y	NA	Y	Y	Y	Y	Y	Y	NA	NA	Y	N	Y	NA	N	**27**
Beattie et al., 2002 (114)	Observentional	Y	Y	Y	Y	N	Y	Y	N	Y	Y	N	N	Y	Y	N	N	N	N	Y	N	Y	Y	Y	N	Y	Y	NA	NA	NA	Y	N	Y	N	N	**21**
Tsiotos et al., 1998 (115)	Descriptive	Y	Y	Y	Y	N	Y	Y	N	Y	Y	N	N	Y	Y	N	Y	N	N	Y	N	Y	Y	Y	N	Y	Y	NA	NA	NA	Y	N	Y	N	N	**22**
Davies et al., 2006 (116)	Descriptive	Y	Y	Y	Y	N	Y	Y	N	Y	Y	N	N	Y	Y	N	N	N	N	Y	N	Y	Y	Y	N	Y	Y	NA	NA	NA	Y	N	Y	N	N	**21**

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Sivasankar et al., 2006 (117)	Descriptive	Y	Y	Y	Y	N	Y	NA	NA	N	N	NA	NA	Y	N	NA	N	NA	NA	Y	Y	NA	Y	Y	Y	Y	Y	Y	NA	NA	Y	N	Y	NA	Y	**28**
van der Wiel et al., 2020 (118)	case series	Y	Y	Y	Y	N	Y	NA	NA	N	N	NA	NA	Y	N	NA	N	NA	NA	Y	Y	NA	Y	Y	Y	Y	Y	NA	NA	NA	Y	N	Y	NA	N	**27**
Luckhurst et al., 2020 (119)	cohort	Y	Y	Y	Y	N	Y	Y	Y	Y	Y	N	N	Y	Y	Y	N	N	Y	Y	N	N	Y	Y	N	Y	Y	NA	NA	NA	Y	N	Y	N	N	**23**
Bartholdy et al., 2020 (120)	Descriptive	Y	Y	Y	Y	N	Y	Y	N	Y	Y	N	N	Y	Y	N		N	N	Y	N	Y	Y	Y	N	Y	Y	NA	NA	NA	Y	N	Y	N	Y	**23**
Liu et al., 2020 (121)	Descriptive	Y	Y	Y	Y	N	Y	Y	N	Y	Y	N	N	Y	Y	N	N	N	N	Y	N	Y	Y	Y	N	Y	Y	NA	NA	NA	Y	N	Y	N	Y	**22**
Larino-Noia et al., 2020 (122)	Descriptive	Y	Y	Y	Y	N	Y	Y	N	Y	Y	N	N	Y	Y	N	N	N	N	Y	N	Y	Y	Y	N	Y	Y	NA	NA	NA	Y	N		N	Y	**22**
Husu et al., 2020 (123)	cohort	Y	Y	Y	Y	N	Y	Y	Y	Y	Y	N	N	Y	Y		N	N		Y	N	N	Y	Y	N	Y	Y	NA	NA	NA	Y	Y	Y	N	N	**24**
Cao et al., 2020 (168)	Descriptive	Y	Y	Y	Y	N	Y	Y	N	Y	Y	N	N	Y	Y	N	N	N	N	Y	N	Y	Y	Y	N	Y	Y	NA	NA	NA	Y	Y	Y	N	Y	**23**
Pandit et al., 2020 (124)	Descriptive	Y	Y	Y	Y	N	Y	Y	N	Y	Y	N	N	Y	Y	NA	N	N	N	Y	N	N	Y	Y	N	Y	Y	NA	NA	NA	Y	Y	Y	N	N	**22**
Carter et al., 2000 (125)	case series	Y	Y	Y	Y	N	Y	NA	NA	N	N	NA	NA	Y	N	NA	N	NA	NA	Y	Y	NA	Y	Y	Y	Y	Y	NA	NA	NA	Y	N	Y	NA	N	**27**
Moyer et al., 2019 (126)	case series	Y	Y	Y	Y	N	Y	NA	NA	Y	Y	NA	NA	Y	N	NA	N	NA	NA	Y	Y	NA	Y	Y	Y	Y	Y	NA	NA	NA	Y	N	Y	NA	N	**29**
Jurgensen et	case series	Y	Y	Y	Y	N	Y	NA	NA	Y	Y	NA	NA	Y	N	NA	N	NA	NA	Y	Y	NA	Y	Y	Y	Y	Y	NA	NA	NA	Y	N	Y	NA	N	**29**

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al., 2017 (127)																																				
Goenka et al., 2018 (128)	Descriptive	Y	Y	Y	Y	N	Y	Y	N	Y	Y	N	N	Y	Y	N	Y	N	N	Y	N	Y	Y	Y	N	Y	Y	NA	NA	NA	Y	N	Y	N	N	**22**
Nabi et al., 2017 (129)	case series	Y	Y	Y	Y	N	Y	NA	NA	N	N	NA	NA	Y	N	N	N	NA	NA	Y	Y	NA	Y	Y	Y	Y	Y	NA	NA	NA	Y	N	Y	NA	N	**26**
Anderloni et al., 2020 (130)	Descriptive	Y	Y	Y	Y	N	Y	Y	N	Y	Y	N	N	Y	Y	N	N	N	Y	N	Y	N	Y	Y	N	Y	Y	NA	NA	NA	Y	N	Y	N		**22**
Wang et al., 2018 (131)	case series	Y	Y	Y	Y	N	Y	NA	NA	N	N	NA	NA	Y	N	N	N	NA	NA	Y	Y	NA	Y	Y	Y	Y	Y	NA	NA	NA	Y	Y	Y	NA	N	**27**
Ausania et al., 2020 (170)	Descriptive	Y	Y	Y	Y	N	Y	Y	N	Y	Y	N	N	Y	Y	N	Y	N	N	Y	N	Y	Y	Y	N	Y	Y	NA	NA	NA	Y	Y	Y	N	N	**23**
Ji et al., 2018 (132)	Descriptive	Y	Y	Y	Y	N	Y	Y	N	Y	Y	N	N	Y	Y	N	Y	N	N	Y	N	Y	Y	Y	N	Y	Y	NA	NA	NA	Y	Y	Y	N	N	**23**
Chandrasekaran et al., 2015 (133)	cohort	Y	Y	Y	Y	N	Y	Y	Y	Y	Y	N	N	Y	Y	Y	Y	N	NA	Y	N	N	Y	Y	N	Y	Y	NA	NA	NA	Y	Y	Y	N	Y	**26**
Wig et al., 2004 (134)	Descriptive	Y	Y	Y	Y	N	Y	Y	N	Y	Y	N	N	Y	Y	N	N	N	NA	Y	N	N	Y	Y	N	Y	Y	NA	NA	NA	Y	Y	Y	N	N	**22**

Abbreviations: Y: Yes, N: NO, NA: Not Applicable

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Items for STROBE checklist (171)

	Item No.	Recommendation
Title and abstract	1	(*a*) Indicate the study's design with a commonly used term in the title or the abstract
		(*b*) Provide in the abstract an informative and balanced summary of what was done and what was found
Introduction		
Background/rationale	2	Explain the scientific background and rationale for the investigation being reported
Objectives	3	State specific objectives, including any prespecified hypotheses
Methods		
Study design	4	Present key elements of study design early in the paper
Setting	5	Describe the setting, locations, and relevant dates, including periods of recruitment, exposure, follow-up, and data collection
Participants	6	(*a*) *Cohort study*—Give the eligibility criteria, and the sources and methods of selection of participants. Describe methods of follow-up *Case-control study*—Give the eligibility criteria, and the sources and methods of case ascertainment and control selection. Give the rationale for the choice of cases and controls *Cross-sectional study*—Give the eligibility criteria, and the sources and methods of selection of participants
		(*b*) *Cohort study*—For matched studies, give matching criteria and number of exposed and unexposed *Case-control study*—For matched studies, give matching criteria and the number of controls per case
Variables	7	Clearly define all outcomes, exposures, predictors, potential confounders, and effect modifiers. Give diagnostic criteria, if applicable

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Data sources/ measurement	8*	For each variable of interest, give sources of data and details of methods of assessment (measurement). Describe comparability of assessment methods if there is more than one group
Bias	9	Describe any efforts to address potential sources of bias
Study size	10	Explain how the study size was arrived at
Quantitative variables	11	Explain how quantitative variables were handled in the analyses. If applicable, describe which groupings were chosen and why
Statistical methods	12	(*a*) Describe all statistical methods, including those used to control for confounding
		(*b*) Describe any methods used to examine subgroups and interactions
		(*c*) Explain how missing data were addressed
		(*d*) *Cohort study*—If applicable, explain how loss to follow-up was addressed *Case-control study*—If applicable, explain how matching of cases and controls was addressed *Cross-sectional study*—If applicable, describe analytical methods taking account of sampling strategy
		(*e*) Describe any sensitivity analyses
Results		
Participants	13*	(a) Report numbers of individuals at each stage of study—eg numbers potentially eligible, examined for eligibility, confirmed eligible, included in the study, completing follow-up, and analysed
		(b) Give reasons for non-participation at each stage
		(c) Consider use of a flow diagram
Descriptive data	14*	(a) Give characteristics of study participants (eg demographic, clinical, social) and information on exposures and potential confounders
		(b) Indicate number of participants with missing data for each variable of interest

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		(c) *Cohort study*—Summarise follow-up time (eg, average and total amount)
Outcome data	15*	*Cohort study*—Report numbers of outcome events or summary measures over time
		Case-control study—Report numbers in each exposure category, or summary measures of exposure
		Cross-sectional study—Report numbers of outcome events or summary measures
Main results	16	(*a*) Give unadjusted estimates and, if applicable, confounder-adjusted estimates and their precision (eg, 95% confidence interval). Make clear which confounders were adjusted for and why they were included
		(*b*) Report category boundaries when continuous variables were categorized
		(*c*) If relevant, consider translating estimates of relative risk into absolute risk for a meaningful time period
Other analyses	17	Report other analyses done—eg analyses of subgroups and interactions, and sensitivity analyses
Discussion		
Key results	18	Summarise key results with reference to study objectives
Limitations	19	Discuss limitations of the study, taking into account sources of potential bias or imprecision. Discuss both direction and magnitude of any potential bias
Interpretation	20	Give a cautious overall interpretation of results considering objectives, limitations, multiplicity of analyses, results from similar studies, and other relevant evidence
Generalisability	21	Discuss the generalisability (external validity) of the study results
Other information		
Funding	22	Give the source of funding and the role of the funders for the present study and, if applicable, for the original study on which the present article is based

*Give information separately for cases and controls in case-control studies and, if applicable, for exposed and unexposed groups in cohort and cross-sectional studies.

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Table S4. Quality assessment of the included Randomized Clinical Trial studies (CONSORT Checklist)

Citation	Study type	1a	1b	2a	2b	3a	3b	4a	4b	5	6a	6b	7a	7b	8a	8b	9	10	11a	11b	12a	12b	13a	13b	14a	14b	15	16	17a	17b	18	19	20	21	22	23	24	25	Total
Bang et al., 2019 (137)	RCT	N	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	N	N	N	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	33
van Santvoort, et al., 2010 (140)	RCT	N	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	N	Y	N	Y	Y	Y	N	NA	Y	Y	Y	Y	Y	Y	Y	Y	Y	N	Y	Y	N	N	Y	Y	Y	Y	30
van Brunschot et al., 2018 (147)	RCT	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	37
Bang et al., 2019 (44)	RCT	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	37
Voermans et al, 2011 (64)	RCT	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	N	Y	Y	Y	Y	Y	Y	Y	NA	Y	Y	Y	Y	Y	N	Y	Y	Y	Y	Y	Y	Y	N	Y	Y	N	Y	33
Bakker et al., 2012 (158)	RCT	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	N	N	NA	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	N	Y	34
Garg et al,, 2020 (169)	RCT	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	NA	Y	Y	Y	Y	Y	NA	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	37

Abbreviations: Y: Yes, N: NO, NA: Not Applicable

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Items for CONSORT 2010 checklist (172)

Title and abstract

(1a) Identification as a randomized trial in the title

(1b) Structured summary of trial design, methods, results, and conclusions

(2a) Scientific background and explanation of rationale

(2b) Specific objectives or hypotheses

Methods

(3a) Description of trial design (such as parallel, factorial) including allocation ratio

(3b) Important changes to methods after trial commencement (such as eligibility criteria), with reasons

(4a) Eligibility criteria for participants

(4b) Settings and locations where the data were collected

(5) The interventions for each group with sufficient details to allow replication, including how and when they were actually administered

(6a) Completely defined pre-specified primary and secondary outcome measures, including how and when they were assessed

(6b) Any changes to trial outcomes after the trial commenced, with reasons

(7a) How sample size was determined

(7b) When applicable, explanation of any interim analyses and stopping guidelines

(8a) Method used to generate the random allocation sequence

(8b) Type of randomization; details of any restriction (such as blocking and block size)

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(9) Mechanism used to implement the random allocation sequence (such as sequentially numbered containers), describing any steps taken to conceal the sequence until interventions were assigned

(10) Who generated the random allocation sequence, who enrolled participants, and who assigned participants to interventions

(11a) If done, who was blinded after assignment to interventions (for example, participants, care providers, those assessing outcomes) and how

(11b) If relevant, description of the similarity of interventions

(12a) Statistical methods used to compare groups for primary and secondary outcomes

(12b) Methods for additional analyses, such as subgroup analyses and adjusted analyses

Results

(13a) For each group, the numbers of participants who were randomly assigned, received intended treatment, and were analyzed for the primary outcome

(13b) For each group, losses and exclusions after randomization, together with reasons

(14a) Dates defining the periods of recruitment and follow-up

(14b) Why the trial ended or was stopped

(15) A table showing baseline demographic and clinical characteristics for each group

(16) For each group, number of participants (denominator) included in each analysis and whether the analysis was by original assigned groups

17a) For each primary and secondary outcome, results for each group, and the estimated effect size and its precision (such as 95% confidence interval)

17b) For binary outcomes, presentation of both absolute and relative effect sizes is recommended

(18) Results of any other analyses performed, including subgroup analyses and adjusted analyses, distinguishing pre-specified from exploratory

(19) All important harms or unintended effects in each group (for specific guidance see CONSORT for harms)

Discussion

(20) Trial limitations, addressing sources of potential bias, imprecision, and, if relevant, multiplicity of analyses

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(21) Generalizability (external validity, applicability) of the trial findings

(22) Interpretation consistent with results, balancing benefits and harms, and considering other relevant evidence

Other information

23) Registration number and name of trial registry

(24) Where the full trial protocol can be accessed, if available

(25) Sources of funding and other support (such as supply of drugs), role of funders

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Supplementary material

Supplementary References (172 studies included in the meta-analysis):

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Supplementary material

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